Enzymes

UNIT XVII



Learning Outcomes

At the end of each topic, the students should be able to:

- ❑ Know the general characteristics of enzymes in terms of function and in terms of structure
- ❑ Be familiar with the terminology used to classify enzymes based on their composition or the types of reaction they catalyze
- ❑ Know what is active site and enzyme-substrate complex and enzyme's mode of action
- ❑ Recognize the different specificity of enzymes
- ❑ Know the factors that can affect enzyme activity and how
- ❑ Enumerate the functions of cofactors, prosthetic and co-enzymes
- ❑ Distinguish the different types of enzyme inhibition and its applications

Importance of Enzymes

- The most important function of proteins
- Catalysis can be up to 10^{20} increase
- Enzymes play an important role in metabolism, diagnosis, and therapeutics.
  - Level of enzyme in blood are of diagnostic importance e.g. it is a good indicator in disease such as myocardial infarction which are levels of lactate dehydrogenase (LDH) and creatine kinase (CK). After a heart attack, CK shows up more rapidly than LDH in the blood.
  - Enzyme can be used therapeutically such as digestive enzymes.

Definition

- Enzymes are biological catalysts
  - They catalyze nearly all the chemical reactions taking place in the cells of the body.
  - They are **proteins** that increase the rate of reaction by looking for an alternative pathway that has a low energy of activation
- **CHARACTERISTICS**
  - They are **not altered or consumed** during reaction.
  - **Reusable** - they can catalyze other reactions after their specified reaction
  - Enzymes show **specificity** to the reaction they control
  - Enzymes are **sensitive to their environment** so they can be controlled by adjusting the temperature, the pH or the substrate concentration

Structure

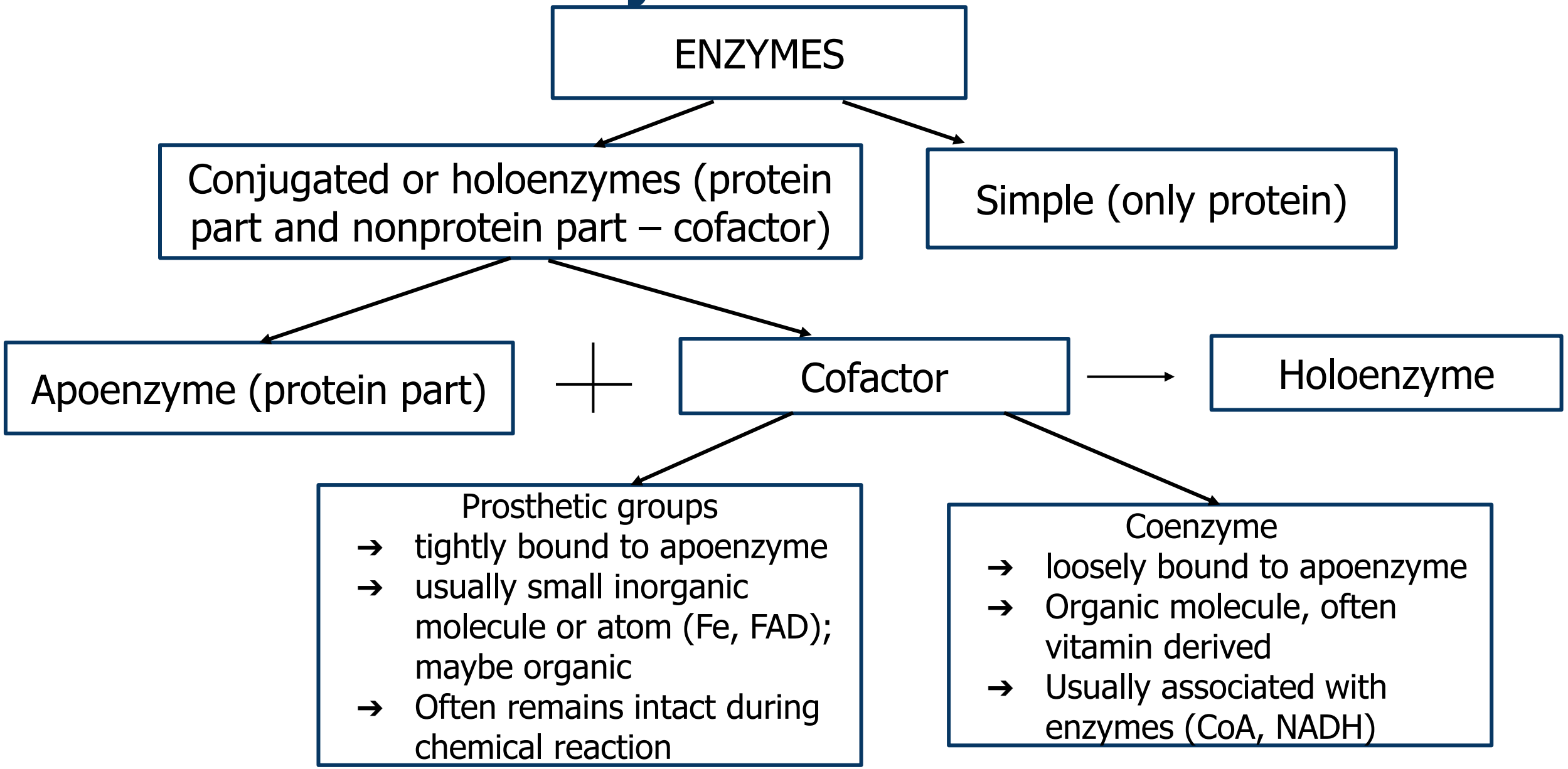
- They have a globular shape
- A complex 3D structure



Human pancreatic amylase

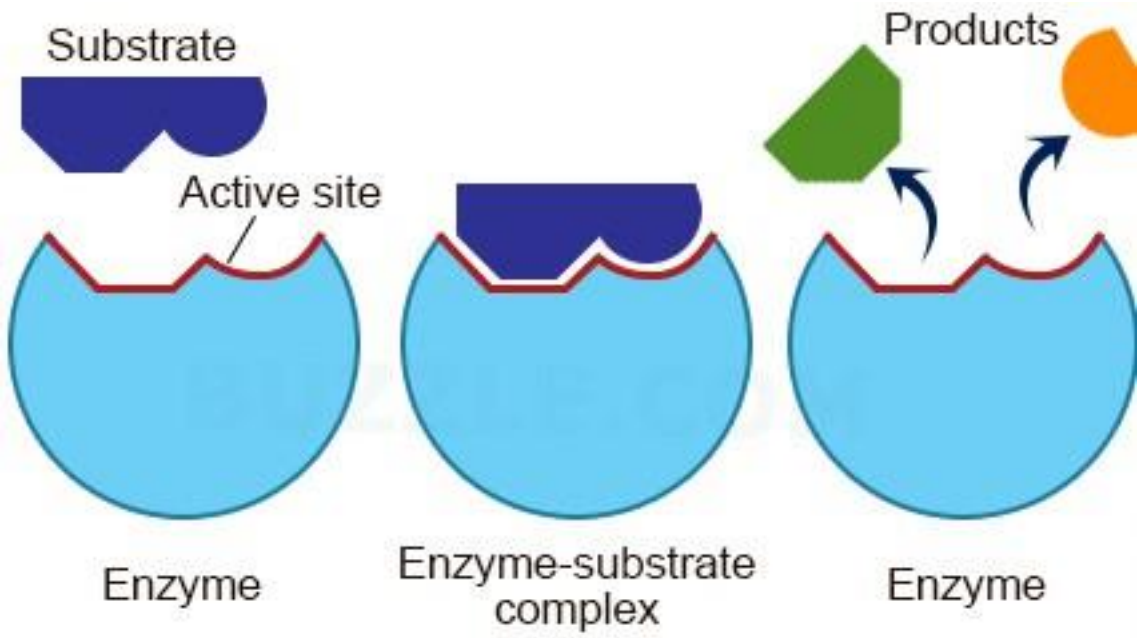
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Structure of Enzymes



Terms to understand biochemical nature of enzymes

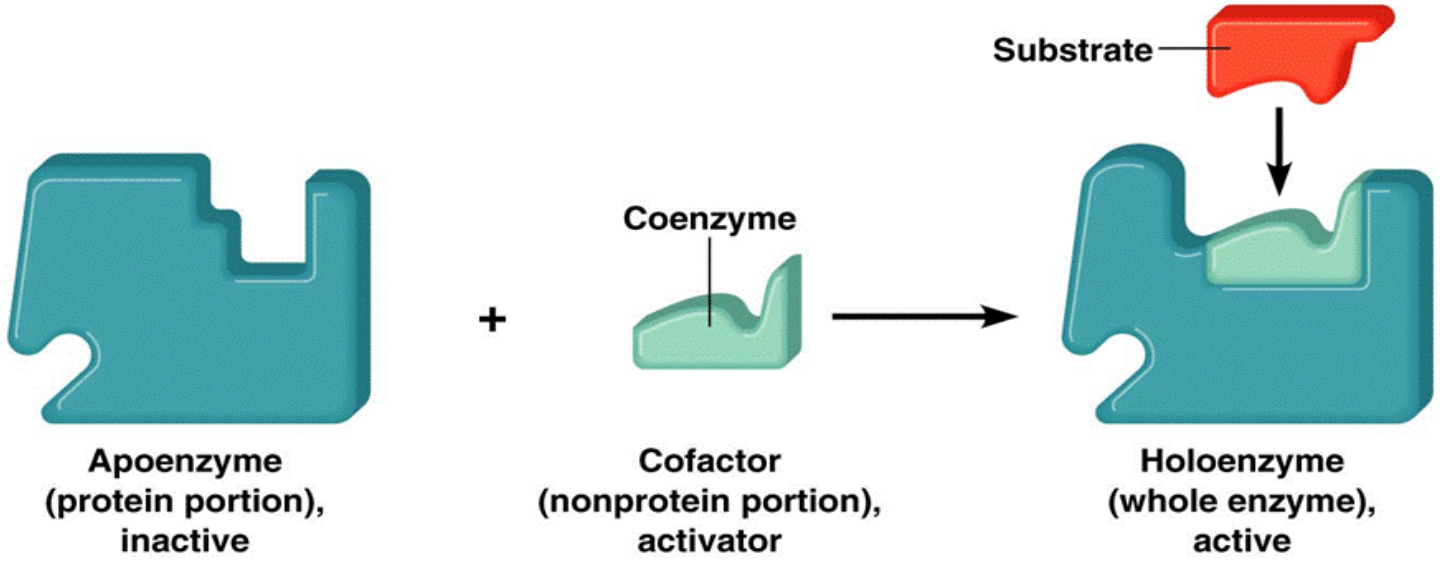
- **Active Site** - The area on the enzyme where the substrate or substrates is/are attached to
  - Enzymes are usually very large proteins and the active site is just a small region of the enzyme molecule.
- **Substrate** – the reactant in the chemical reaction



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Terms to understand biochemical nature of enzymes

- **Apoenzyme** – the **enzyme without its non protein moiety** is termed as apoenzyme and it is *inactive*.
- **Holoenzyme** – the ***active*** **enzyme with its non protein component**.



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Naming Enzymes

- The name of an enzyme in many cases end in –ase
  - For example, *sucrase* catalyzes the hydrolysis of sucrose
  - Sometimes common names are used, particularly for the digestion enzymes. For example, *pepsin* and *trypsin*
- The name describes the function of the enzyme
  - For example, *oxidase* catalyze oxidation reactions
- Some names describe both the substrate and the function
  - For example, *alcohol dehydrogenase* oxides ethanol

Classification of Enzymes

- Enzymes Are Classified into six functional Classes (EC number Classification) by the International Union of Biochemists (I.U.B.) on the Basis of the Types of Reactions that they catalyze

  - EC 1. Oxidoreductases
  - EC 2. Transferases
  - EC 3. Hydrolases
  - EC 4. Lyases
  - EC 5. Isomerases
  - EC 6. Ligases

Principle of International Classification

- Each enzyme is assigned an **Enzyme Commission (EC)** number consisting of four numbers separated by periods.

  Example: EC 2.7.1.1 – Hexokinase (Common Name)

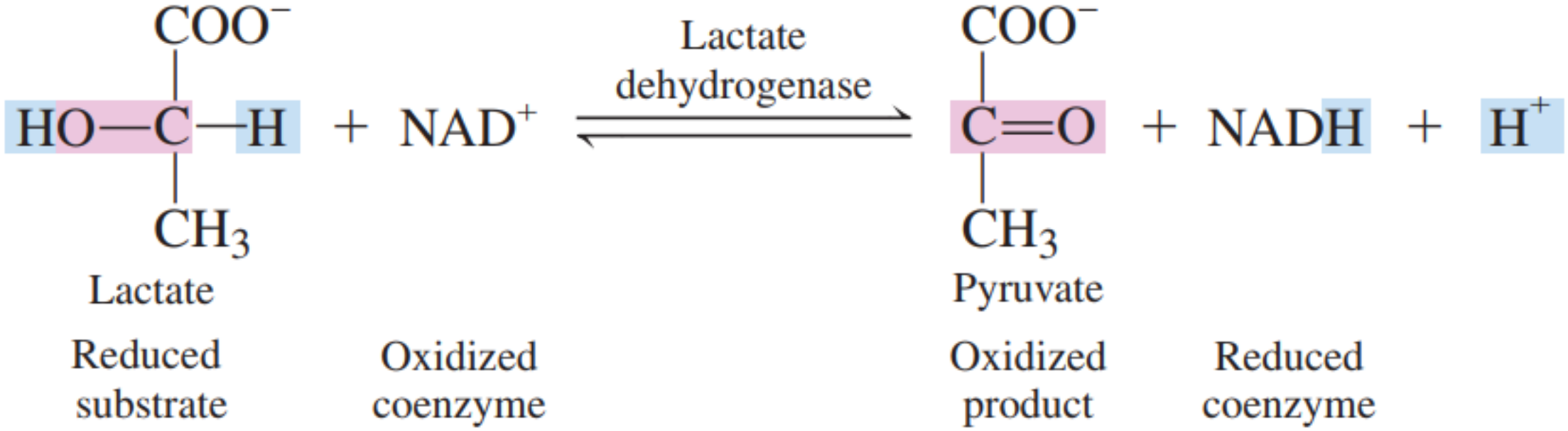
  - 2 - CLASS (Transferases)
  - 7 - SUBCLASS (Transfer of phosphorus-containing group)
  - 1 - SUB-SUBCLASS (Phosphotransferases with an alcohol group as acceptor)
  - 1 – SERIAL NUMBER: Specific enzyme identification

- Systematic name of Hexokinase is:

  ATP,D-HEXOSE-6-PHOSPHOTRANSFERASE

EC 1. Oxidoreductases

- Catalyze *oxidation-reduction* reactions

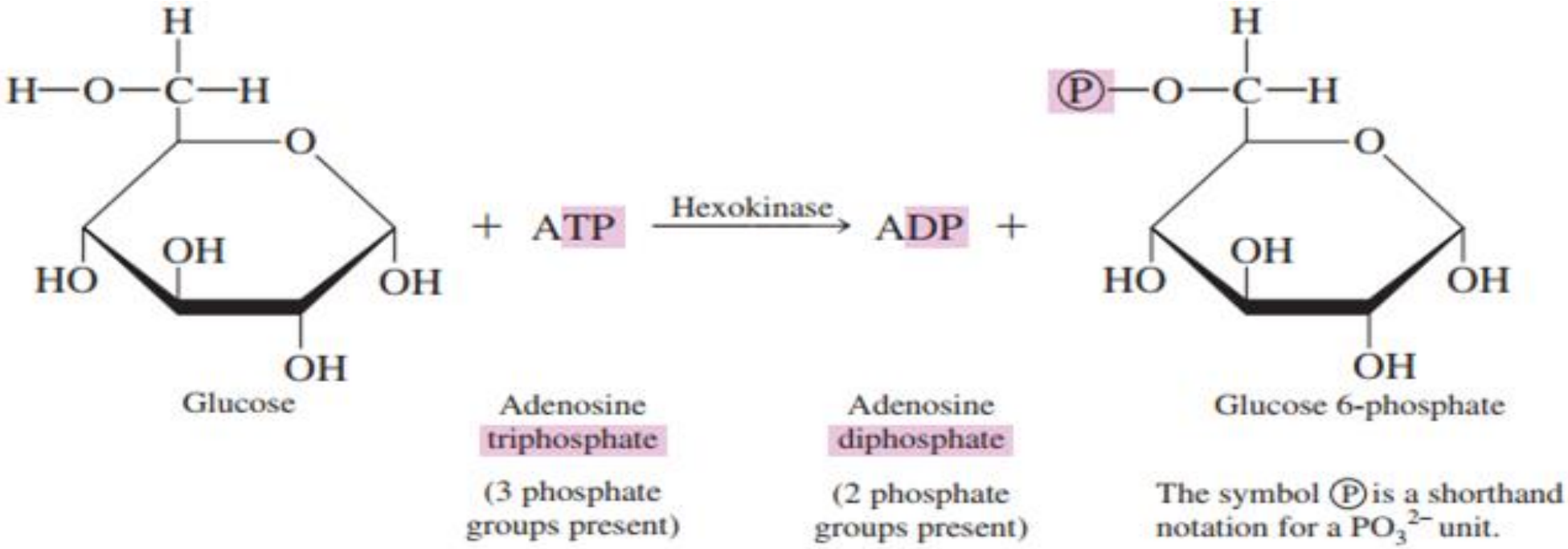


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- Examples: Glucose Oxidase, Peroxidase, Catalase, Phenylalanine hydroxylase

EC 2. Transferases

- Catalyze group transfer reactions

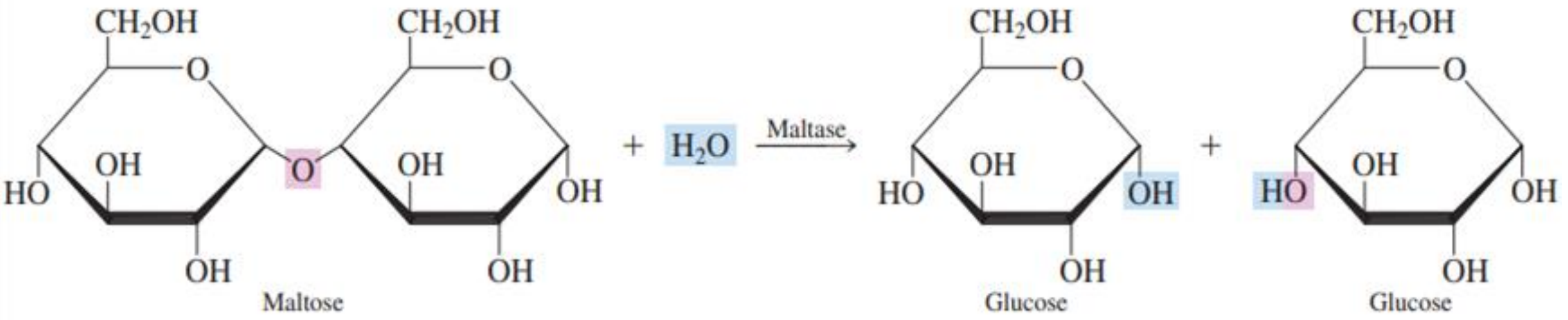


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Examples: Transaminases (ALT & AST), Phosphotransferases (Kinases), Transmethylases, Transpeptidases, Transacylases

EC 3. Hydrolases

- Catalyze hydrolysis reactions where water is the acceptor of the transferred group

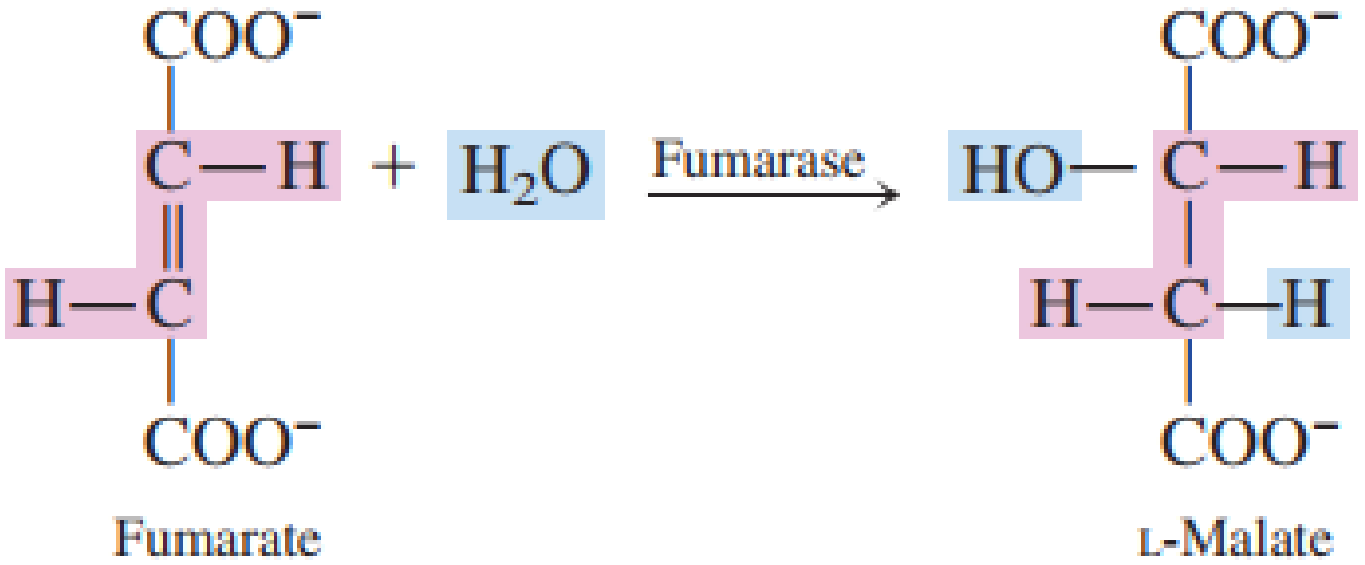


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Other examples: Protein hydrolyzing enzymes (Peptidases), Carbohydrases (Amylase, Maltase, Lactase), Lipid hydrolyzing enzymes (Lipase), Deaminases, Phosphatases

EC 4. Lyases

- Cleave various bonds by means other than hydrolysis and oxidation.
- Add Water, Ammonia or Carbon dioxide across double bonds, or remove these elements to produce double bonds.

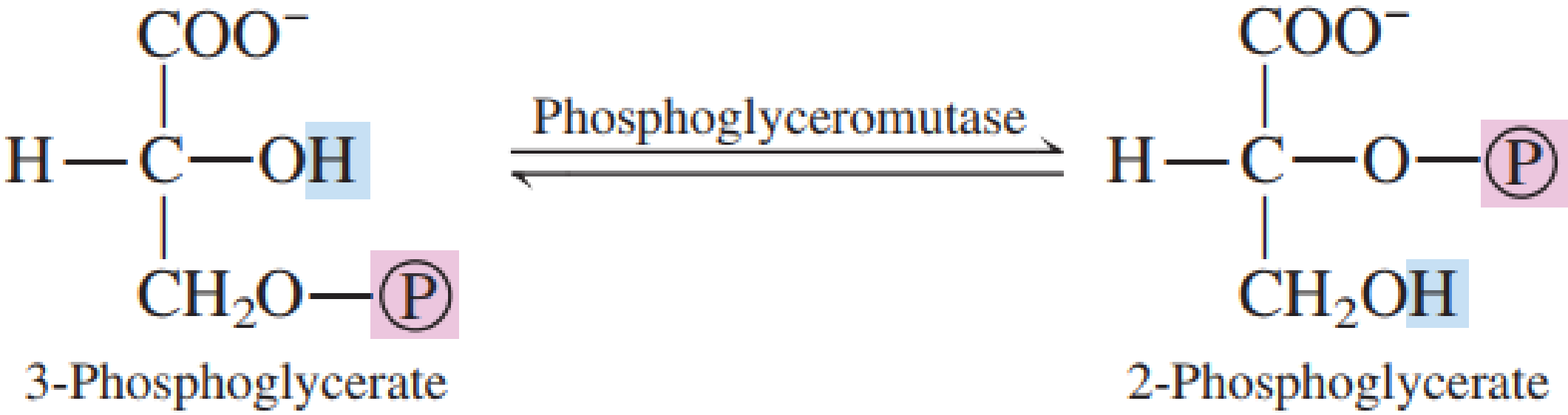


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Examples: Fumarase, Carbonic anhydrase. Hydratase, Dehydratase

EC 5. Isomerase

- Catalyze isomerization reactions



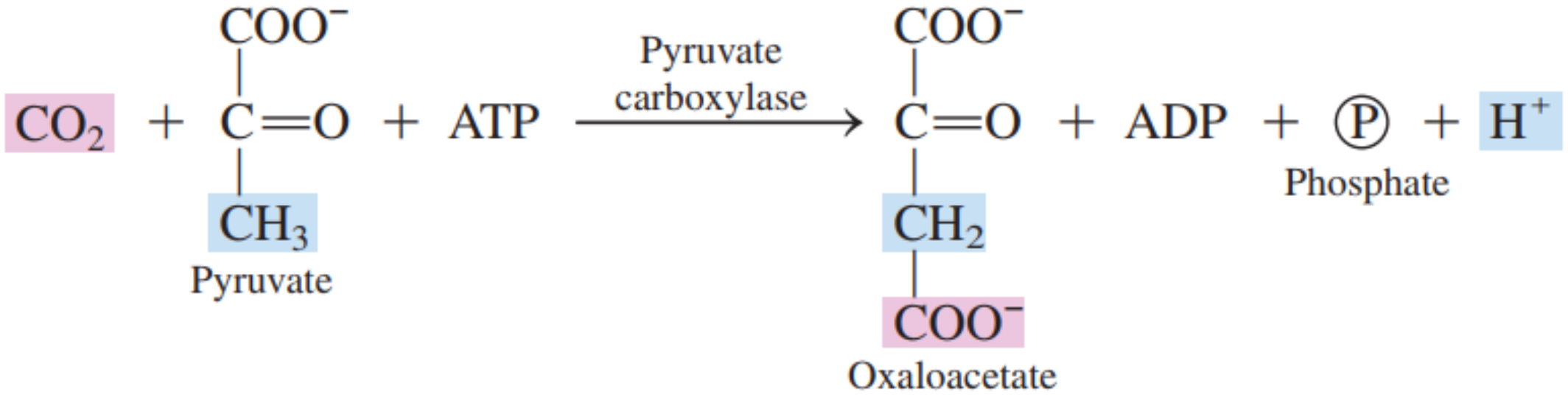
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- Carry out many kinds of isomerization:
  - L to D isomerizations.
  - Mutase reactions (Shifts of chemical groups).

Examples: Isomerase, Mutase

EC 6. Ligases

- Join two molecules with covalent bonds
- Catalyse reactions in which two chemical groups are joined (or ligated) with the use of energy from ATP.



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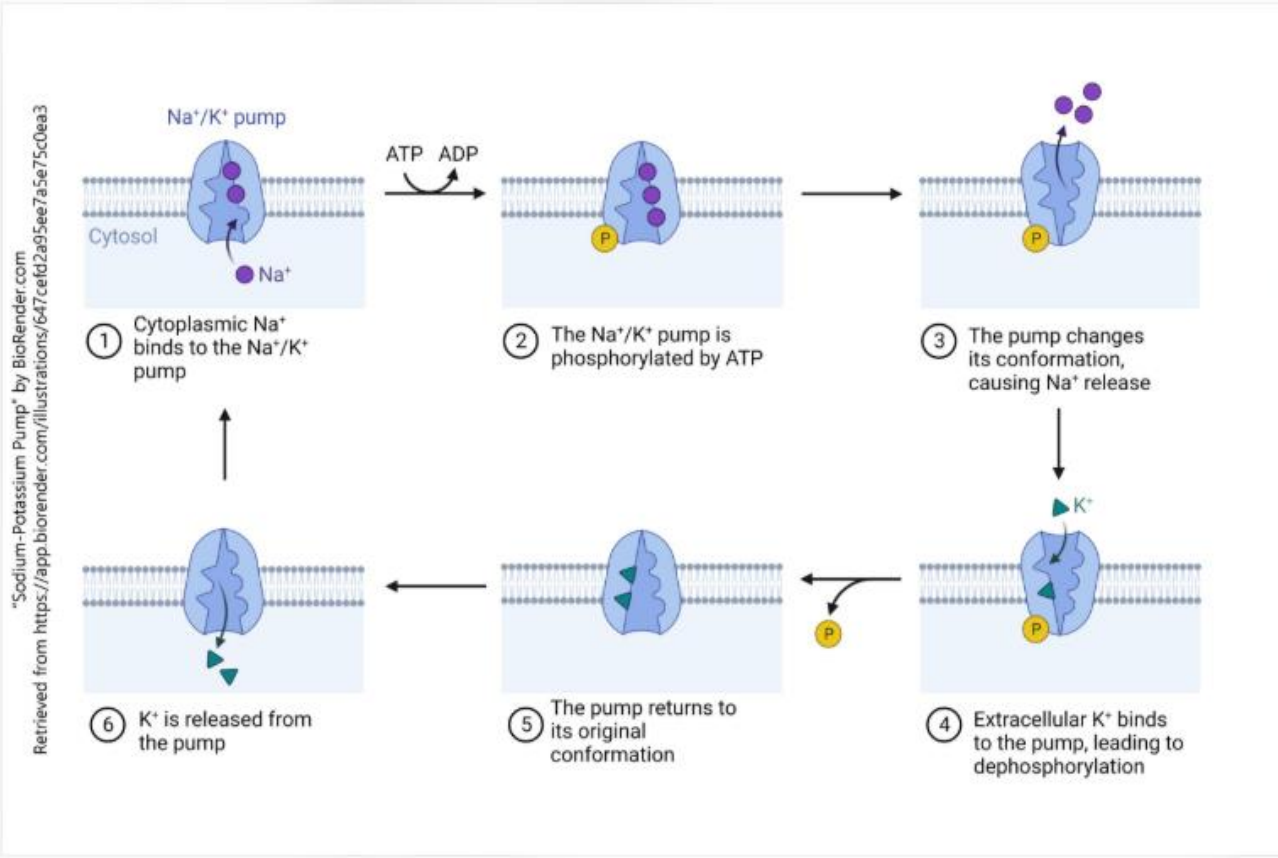
Examples: Acetyl-CoA Carboxylase, Glutamine synthetase

EC 7. Translocases

- Catalyze the translocation of ions or molecules across membranes.
- Transport ions or molecules across biological membranes.
- Important in maintaining electrochemical gradients and cellular transport.

$$3Na^{+}_{in} + 2K^{+}_{out} + ATP + H_2O$$

$$\longrightarrow 3Na^{+}_{out} + 2K^{+}_{in} + ADP + P_i$$

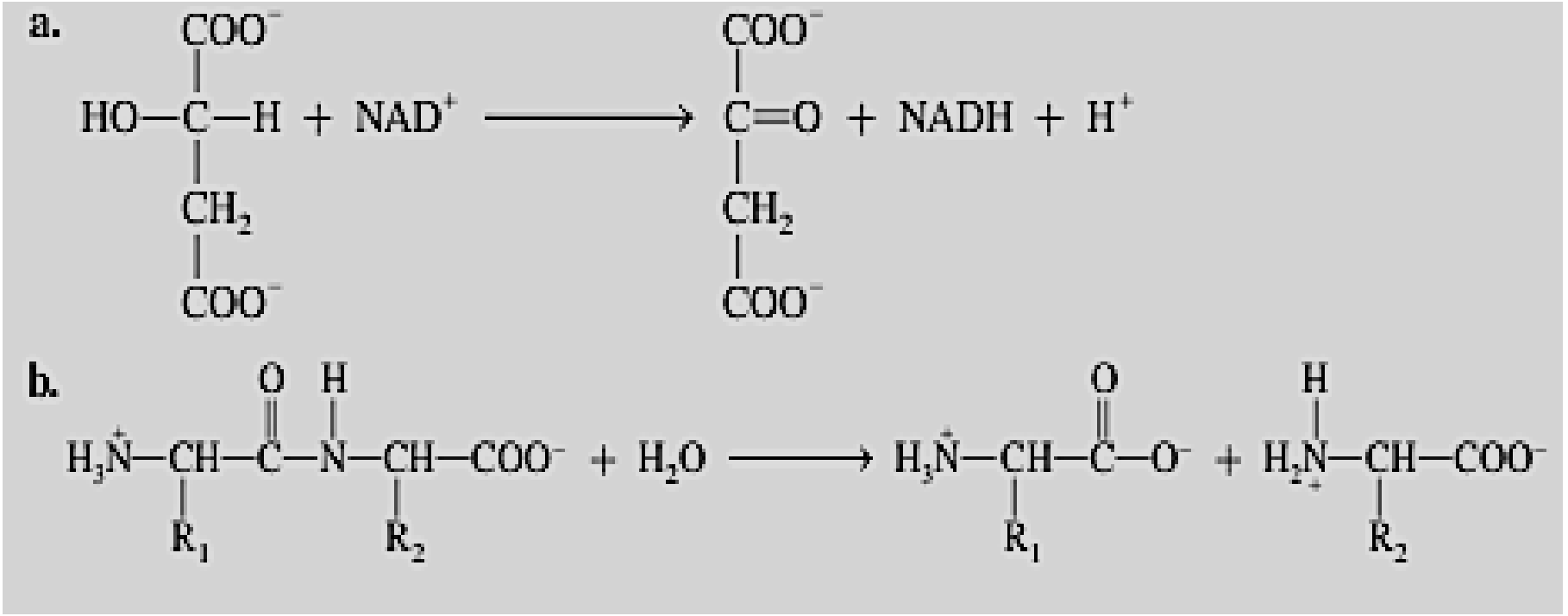


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EXERCISE

Classifying Enzymes by the Type of Chemical Reaction They Catalyze

To what main enzyme class do the enzymes that catalyze the following chemical reactions belong?

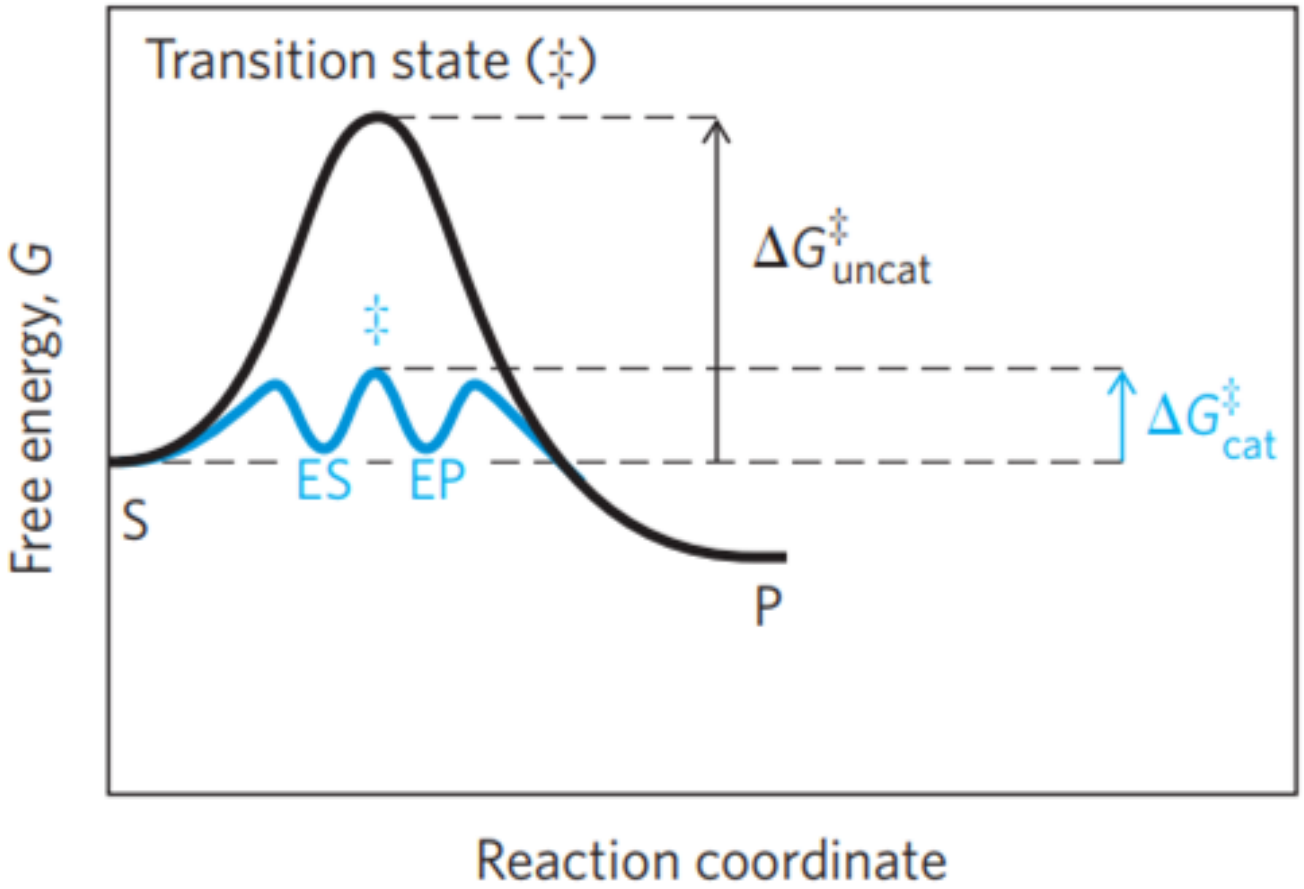


Enzymatic Action

- Chemical reactions need an initial input of energy
  = THE ACTIVATION ENERGY
- During this part of the reaction the molecules are said to be in a **Transition state.**

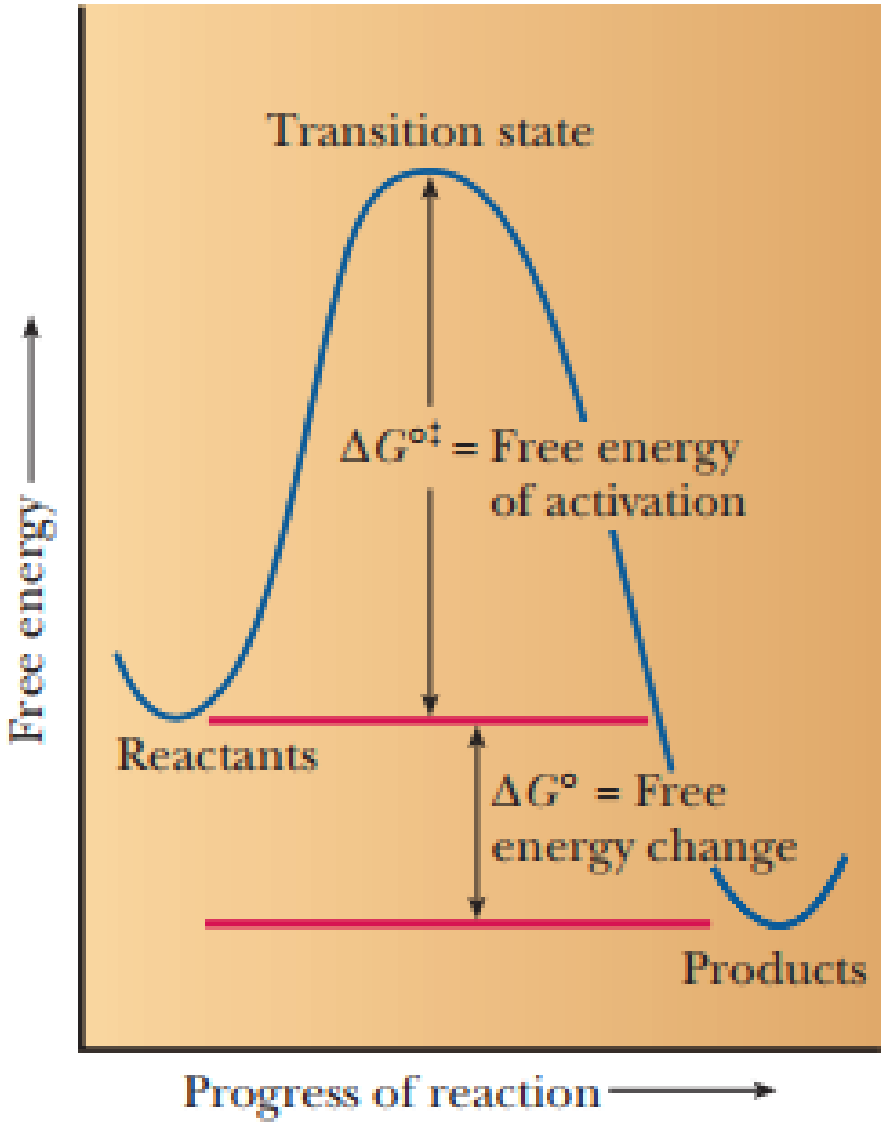
Figure of Reaction Pathwa

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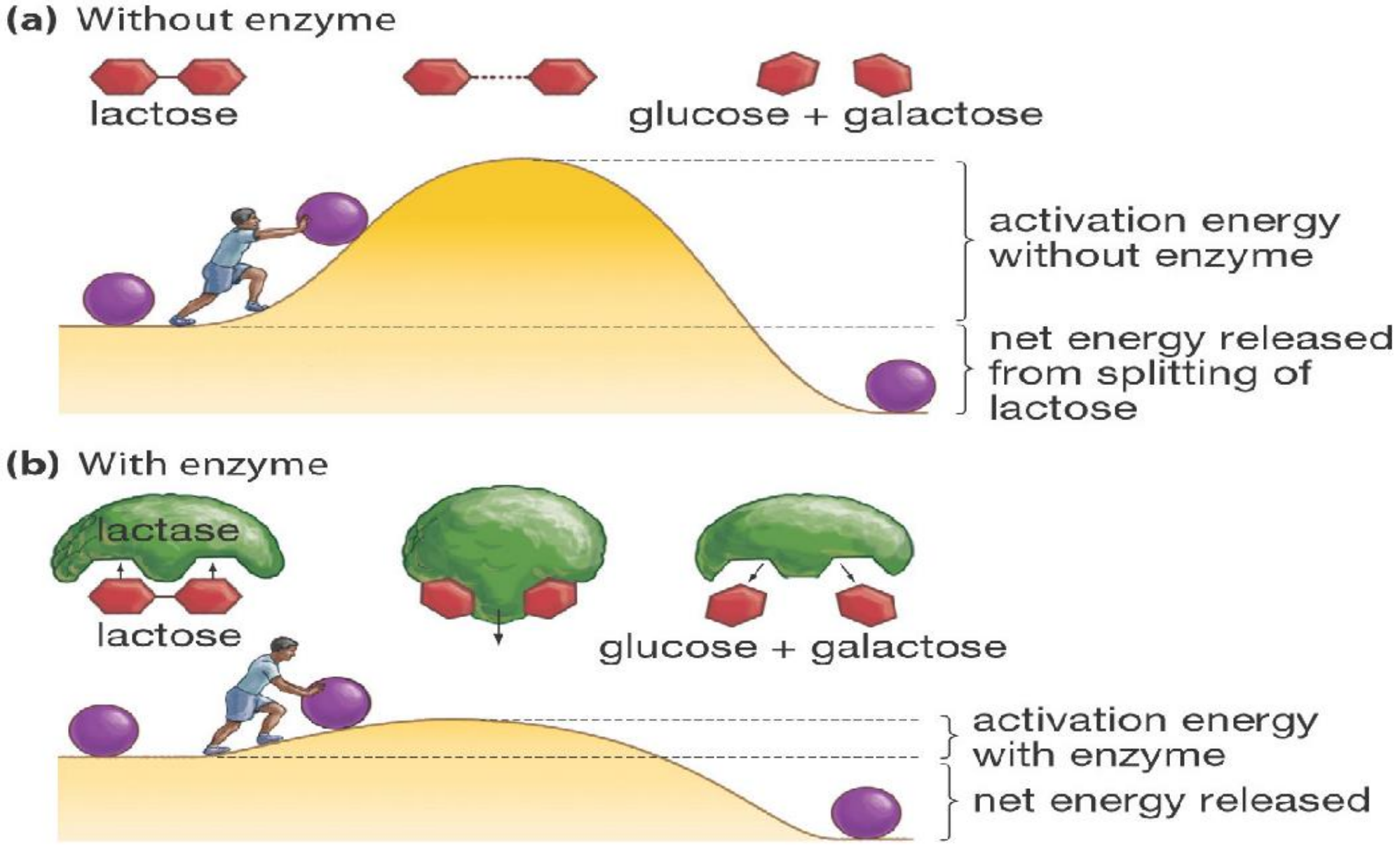
Enzymatic Action

- Biological systems are very sensitive to temperature changes.
- Enzymes can increase the rate of reactions without increasing the temperature.
- They do this by lowering the activation energy. They look for a new reaction pathway "a short cut"
- Enzyme controlled reactions proceed 10^8 to 10^{11} times faster than corresponding non-enzymatic reactions.



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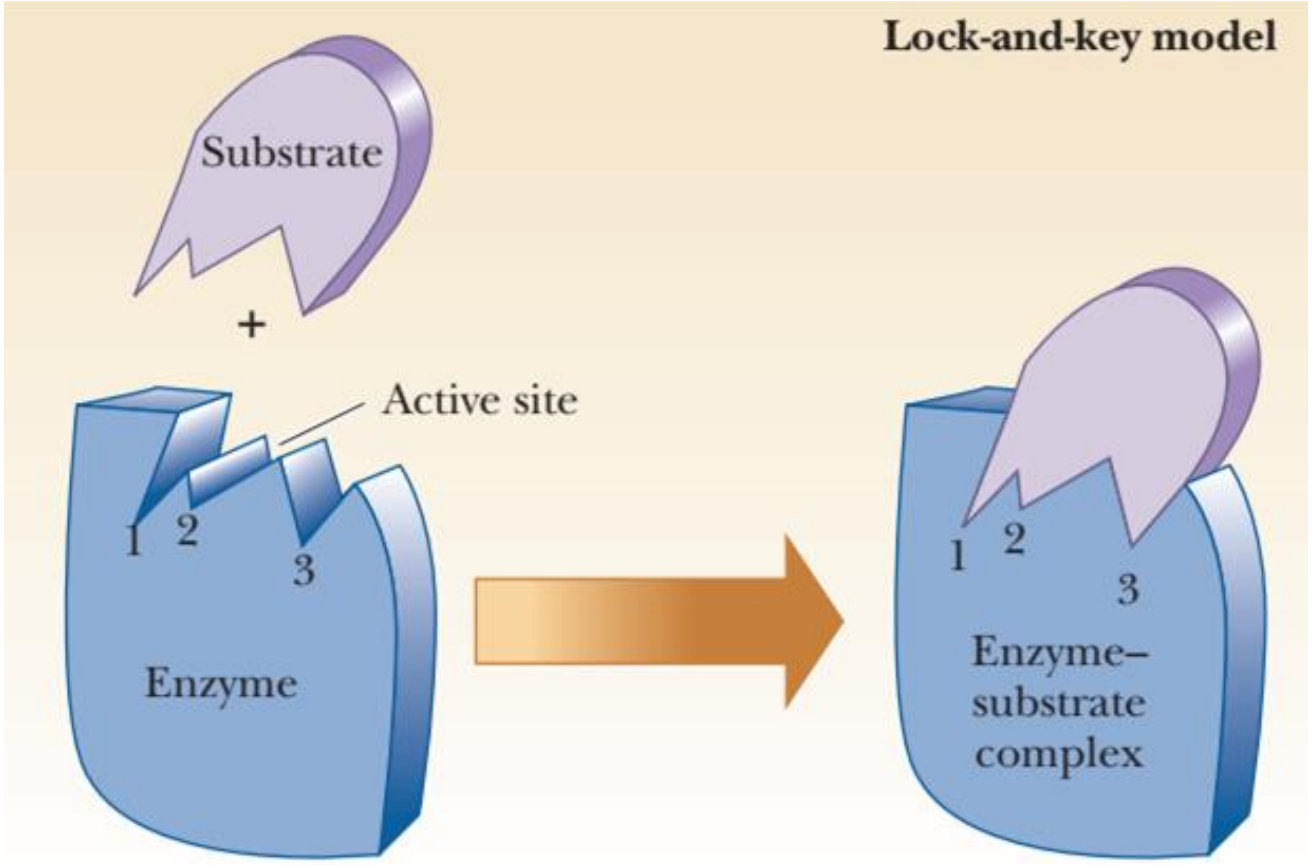
Enzymes Lower a Reaction's Activation Energy



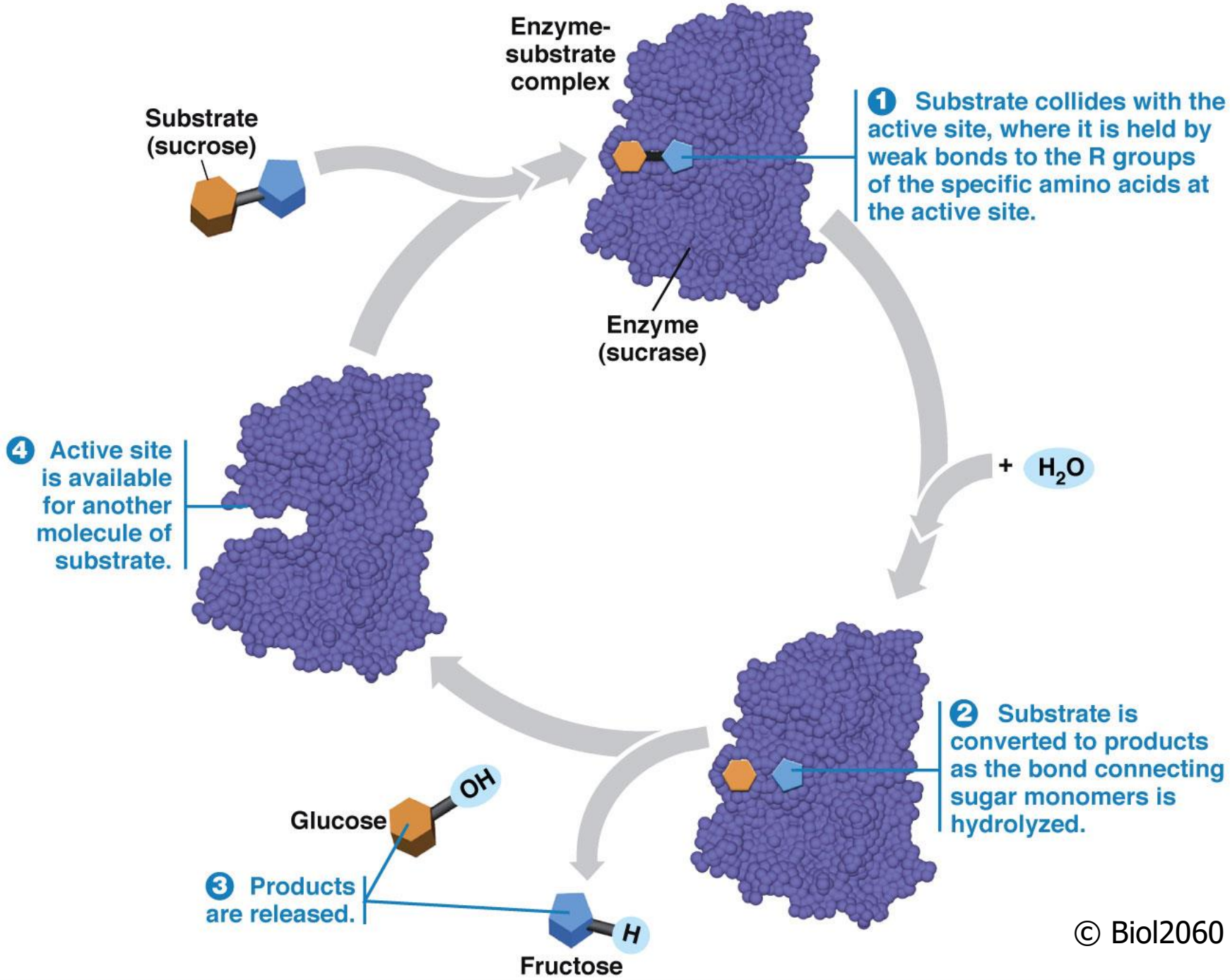
Enzyme-Substrate Binding

TWO MODELS FOR THE ENZYME-SUBSTRATE FORMATION

- **LOCK-AND-KEY MODEL**
  - the active site has a rigid shape
  - only substrates with the matching shape can fit
  - the substrate is a key that fits the lock of the active site



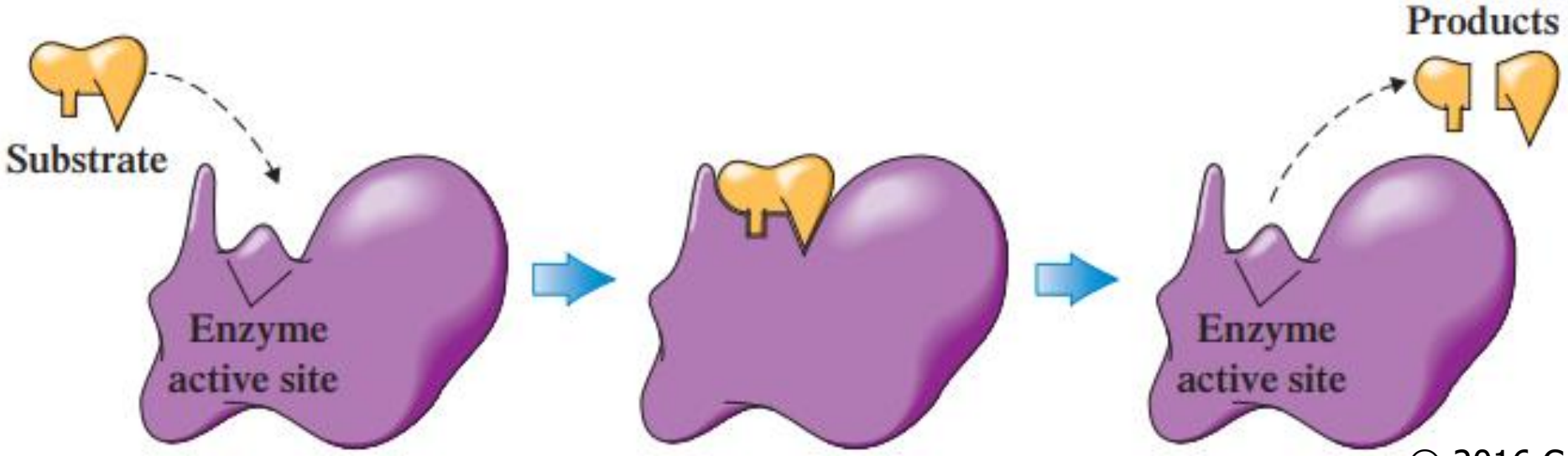
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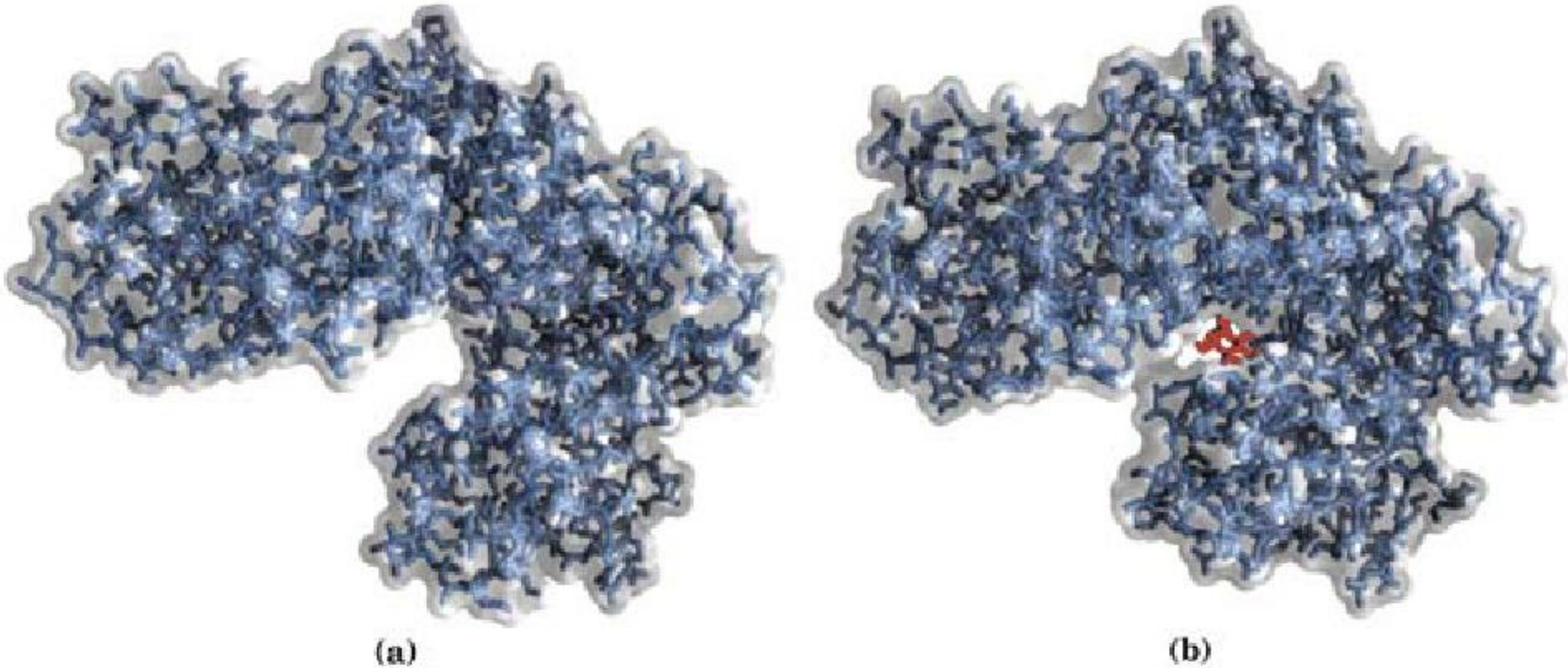
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Enzyme-Substrate Binding

- **INDUCED-FIT MODEL**
  - the active site is flexible, not rigid. It can change its conformation.
  - the shapes of the enzyme, active site, and substrate adjust to maximize the fit, which improves catalysis
  - there is a greater range of substrate specificity
  - this model is more consistent with a wider range of enzymes



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Hexokinase (a) without (b) with glucose substrate

http://www.biochem.arizona.edu/classes/bioc462/462a/NOTES/ENZYMES/enzyme_mechanism.html

Enzyme Specificity

Specificity Type	Reaction	Specific Example
Absolute	Catalyze only **one reaction for one specific substrates**	Urease catalyzes only the hydrolysis of urea
Group	Catalyze only one reaction for substrates with **similar functional groups**	- Hexokinase adds a phosphate group to hexoses (glucose, fructose, mannose) - Alcohol dehydrogenase (ADH) – range of alcohols
Linkage	One type of reaction for a **specific type of bond**	- Salivary amylase (hydrolyzes α-1,4-glycosidic bonds in starch)
Stereospecificity	act on a particular stereoisomer. **Chirality** is inherent in an enzyme active site	L-amino acid oxidase will catalyze the oxidation of the L-form of an amino acid but not the D-form

Factors Affecting Enzyme Activity

Environmental and physicochemical factors – influences the rate of reaction

- Temperature
- pH
- Substrate concentration
- Enzyme Concentration

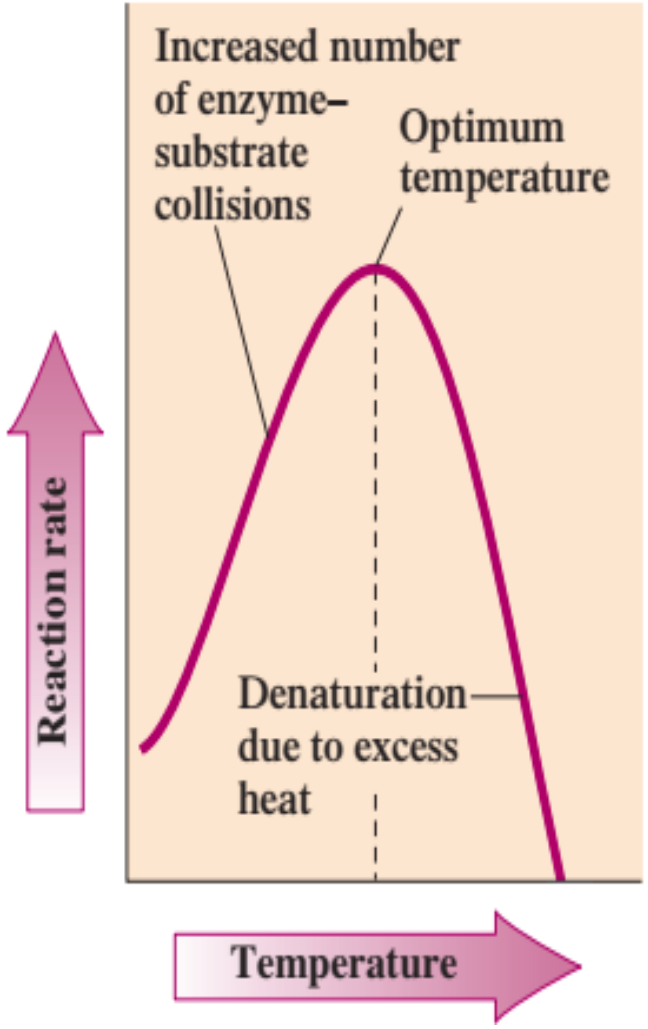
Regulatory molecules and mechanisms

- **Activators**
  - Cofactors
  - Alloesteric activators
- **Inhibitors**
  - Competitive inhibition
  - Noncompetitive inhibition
  - Uncompetitive inhibition
  - Irreversible inhibition
  - Allosteric Inhibition

Factors Affecting Enzyme Activity

Effect of Temperature

- Temperature increases enzyme activity up to its optimum temperature
- **Optimum temperature** is the temperature at which enzymatic reaction occur fastest
- For most enzymes the optimum temperature is about 40°C
- Many are a lot lower, cold water fish will die at 30°C because their enzymes denature
- A few bacteria have enzymes that can withstand very high temperatures up to 100°C
- Most enzymes however are fully denatured at 70°C

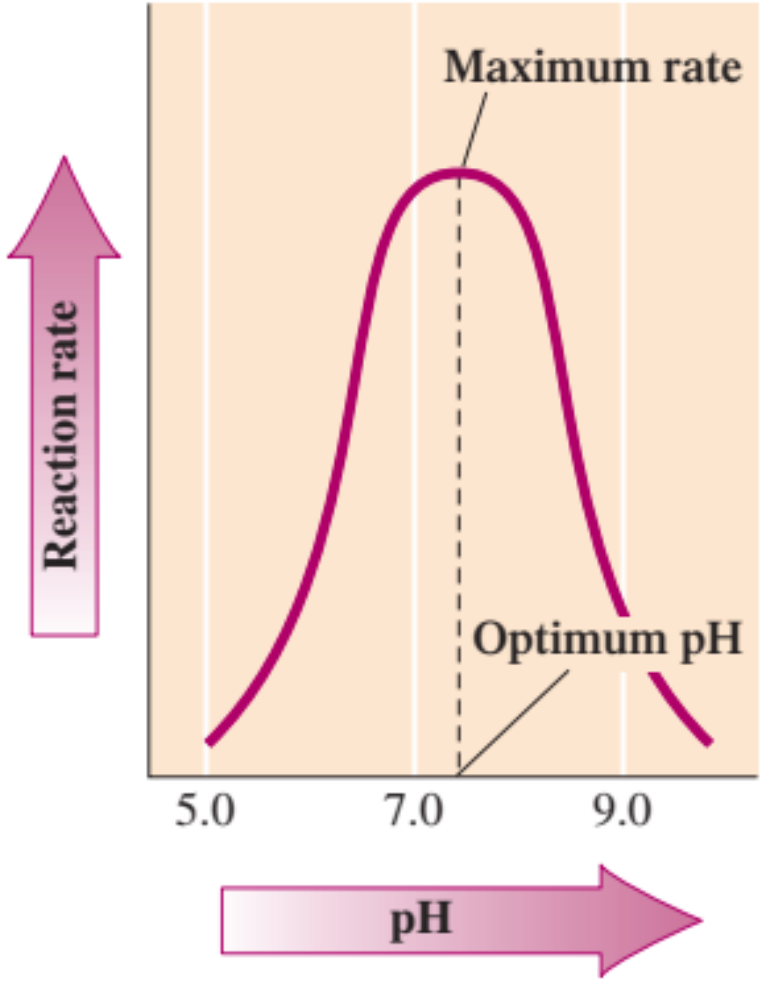


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Factors Affecting Enzyme Activity

Effect of pH

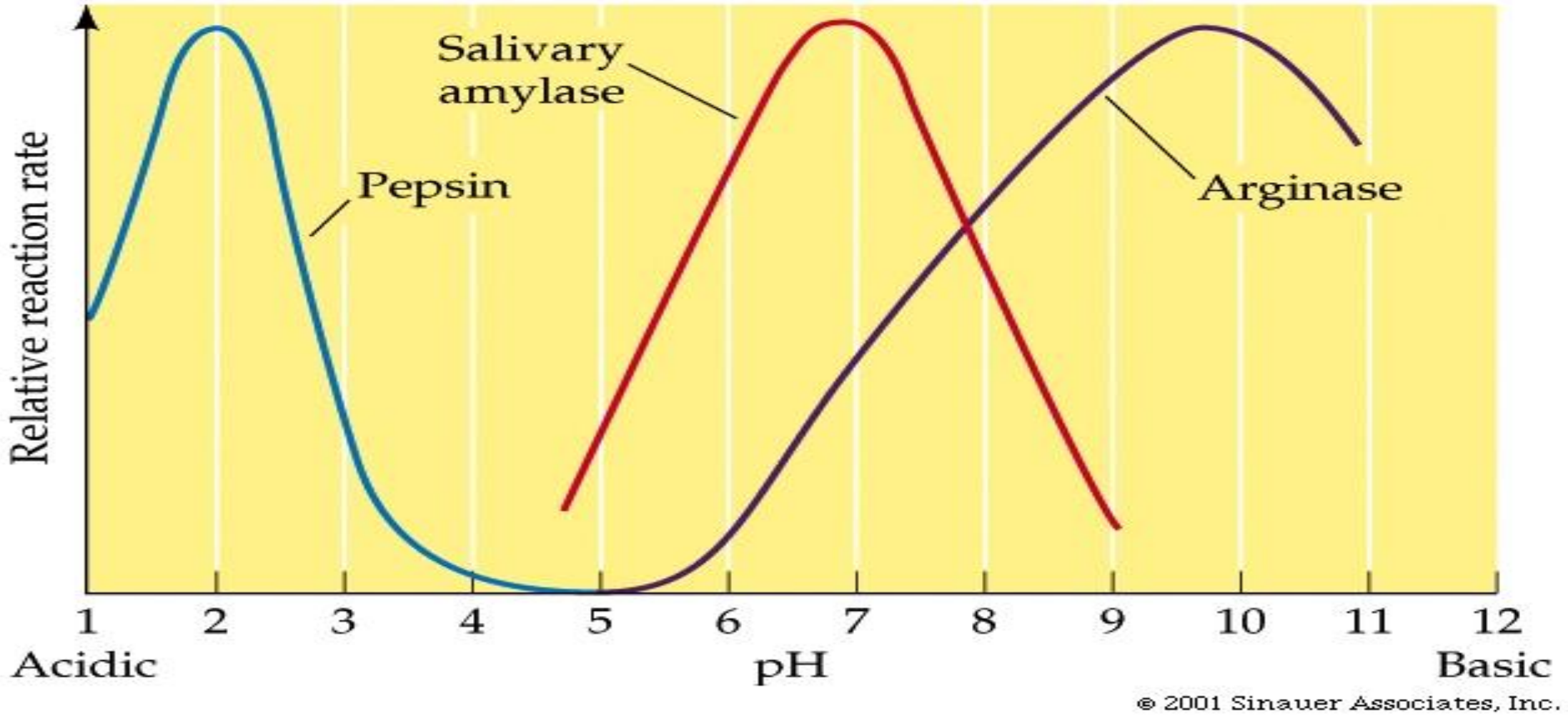
- If pH slightly different from the enzyme's optimum value, there will be small changes in the charges of the enzyme and it's substrate's molecule.
- This change in ionization will affect the binding of the substrate with the active site.
- different optimum pH where enzyme activity is at its highest
- Extreme changes in pH can cause denaturation



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Factors Affecting Enzyme Activity

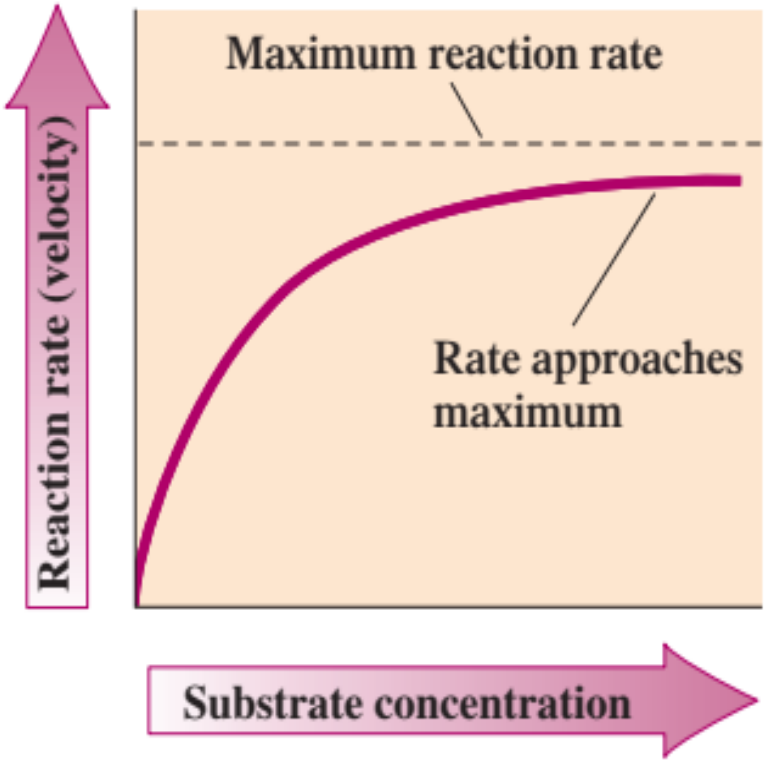
Effect of pH



Factors Affecting Enzyme Activity

Effect of substrate concentration

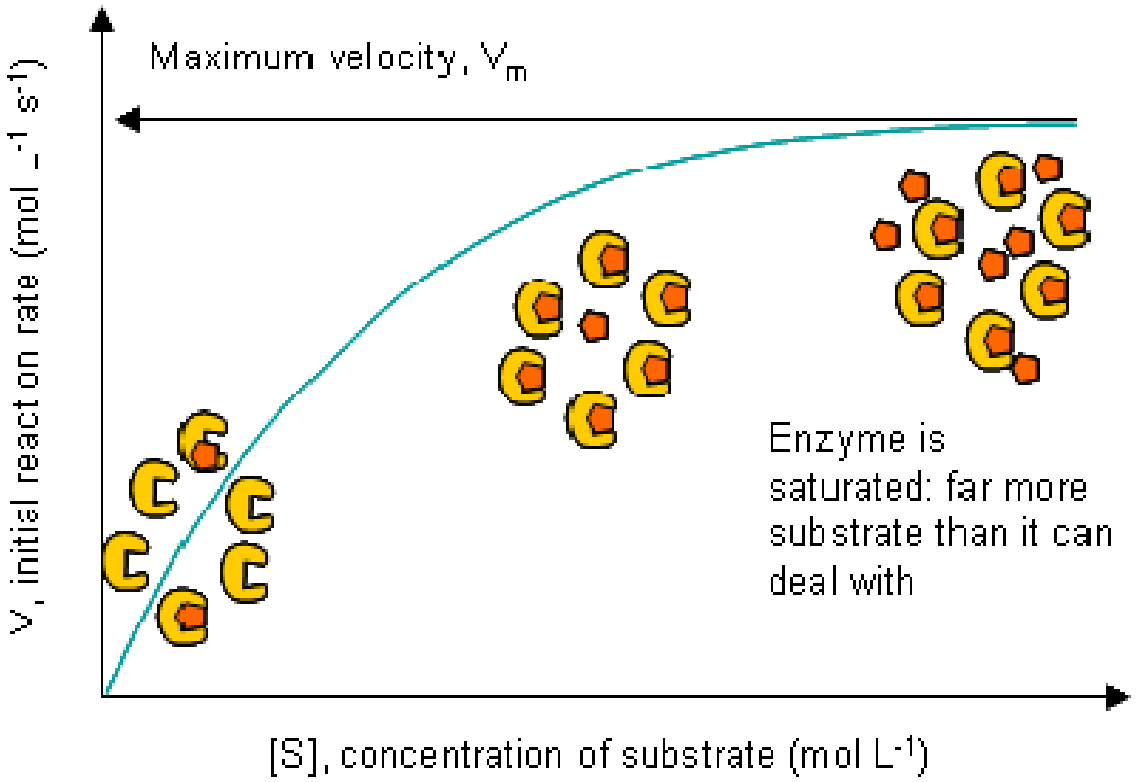
- The rate of reaction increases as substrate concentration increases (at constant enzyme concentration)
- Maximum activity occurs when the enzyme is saturated (when all enzymes are bound to the substrate). This is called **saturation point**.
- If you alter the concentration of the enzyme then Vmax will change too.

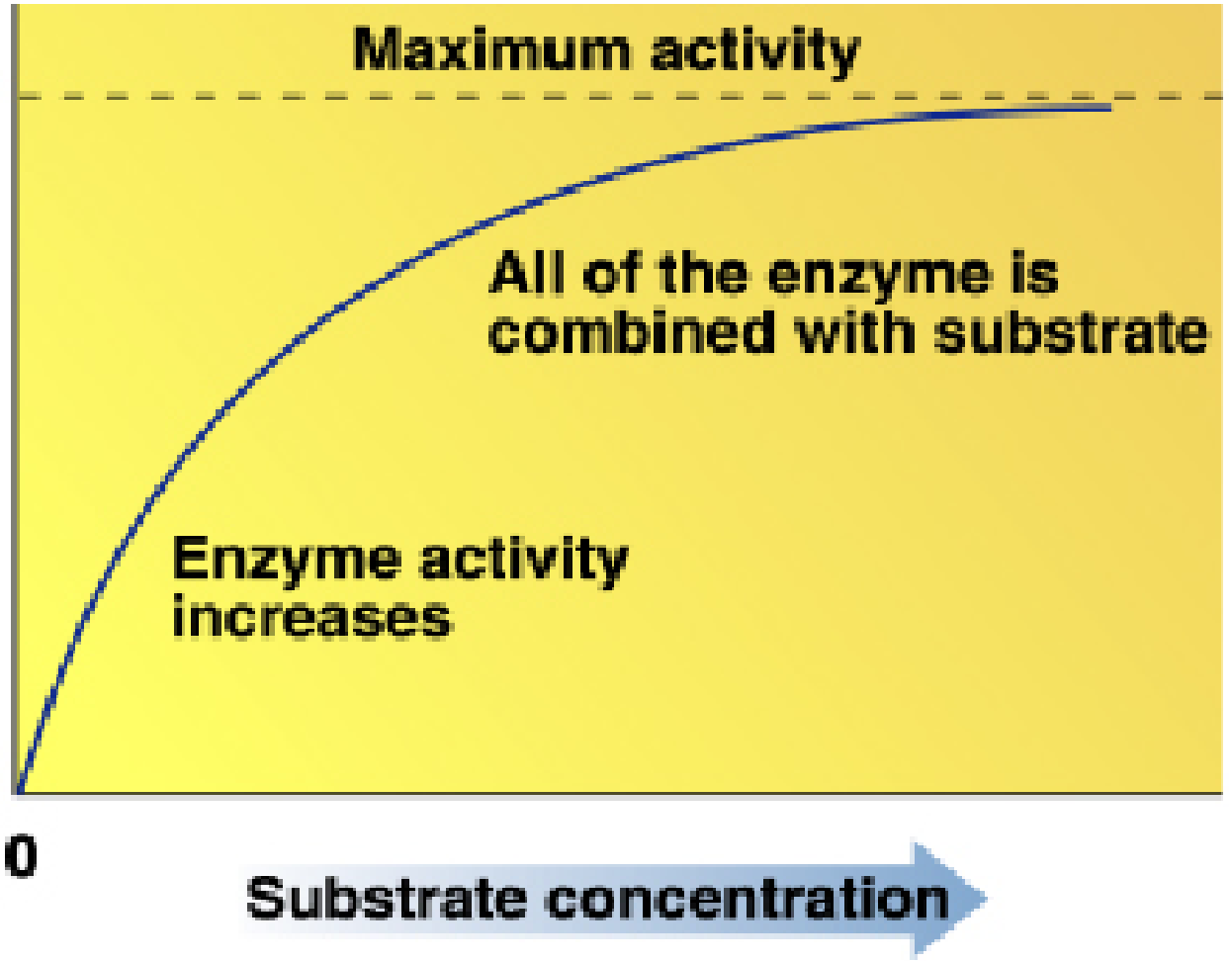


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Factors Affecting Enzyme Activity

Effect of substrate concentration

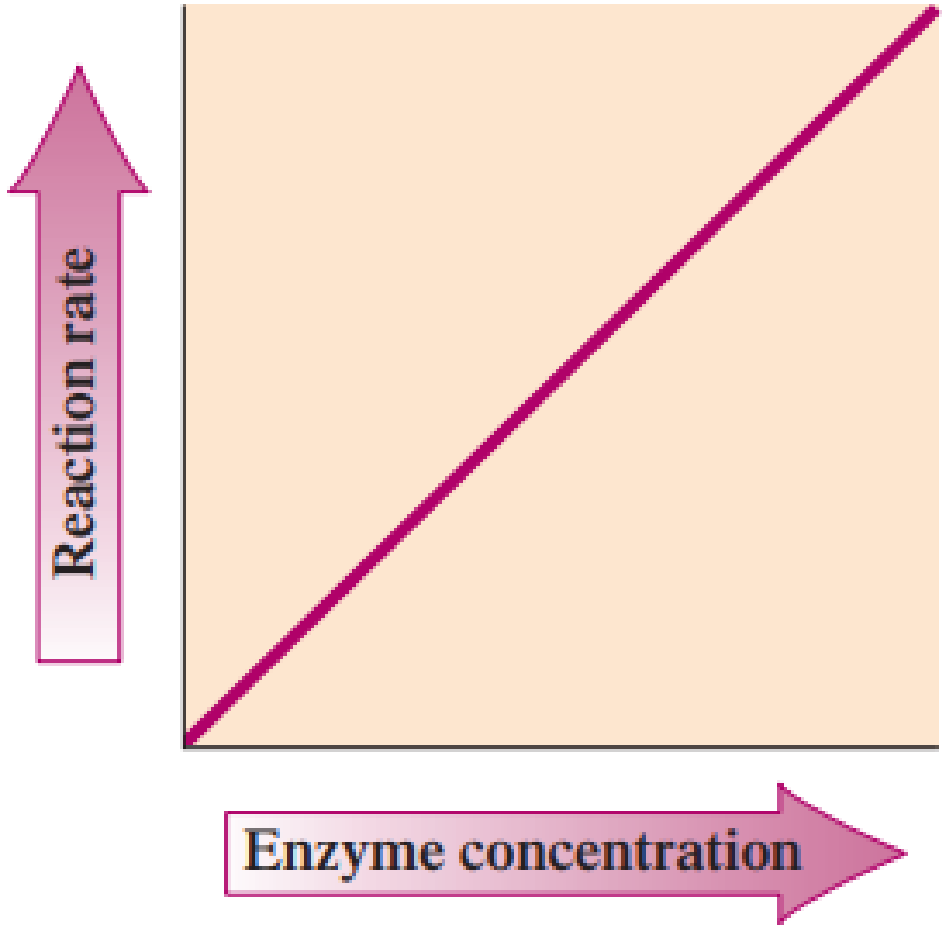




Factors Affecting Enzyme Activity

Effect of enzyme concentration

- Reaction rate increases with increasing enzyme concentration, assuming enzyme concentration is much lower than that of substrate.

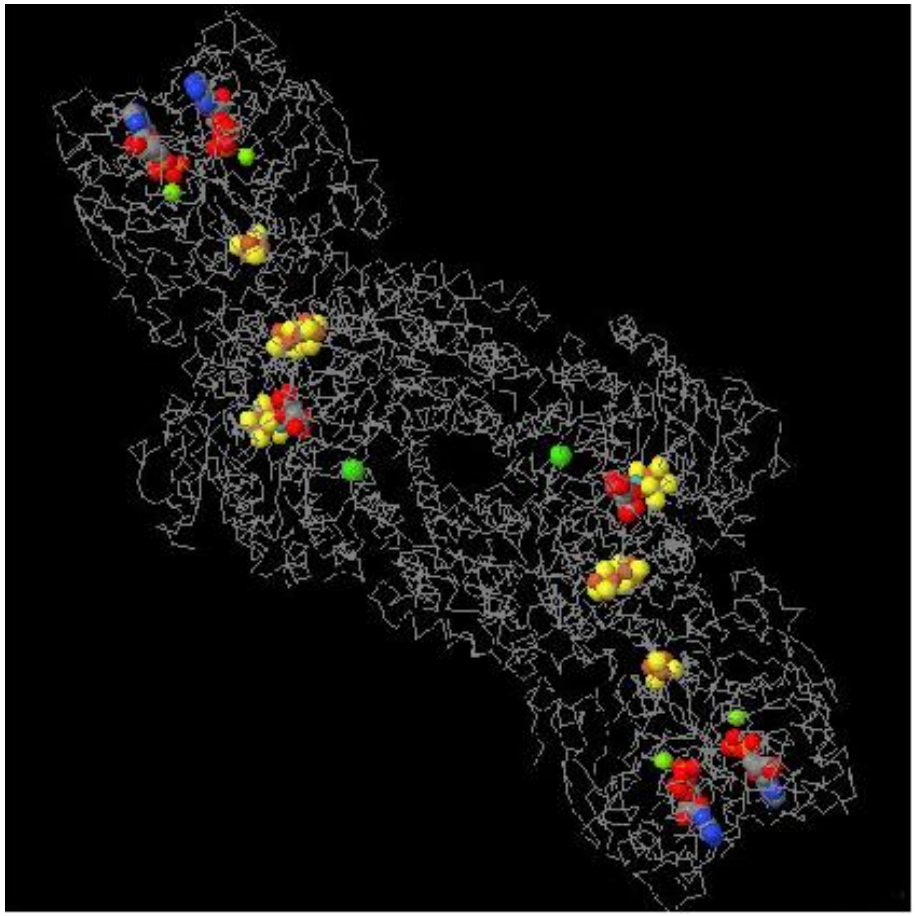


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Factors Affecting Enzyme Activity

Cofactors

- An additional non-protein molecule that is needed by some enzymes necessary for proper functioning. Their main role is to assist.
- Tightly bound cofactors to the apoenzyme are called ***prosthetic groups***
- Cofactors that are bound and released easily to the apoenzyme are called coenzymes.
- Many vitamins are coenzymes. They are not specific to which enzyme they assist.



Nitrogenase enzyme with Fe, Mo and ADP cofactors

Jmol from a RCSB PDB file © 2007 Steve Cook

Factors Affecting Enzyme Activity

Examples of Metal Cofactors

- **Cu^{2+} -** assists cytochrome oxidase, last enzyme in the respiratory electron transport chain
- **Fe^{2+} and Fe^{3+}** - helps catalase, **converts hyrogen peroxide to water and oxygen**.
- **Mg^{2+} -** assists Hexokinase in transferring phosphate to glucose, forming glucose-6-phosphate in glycolysis
- **Mn^{2+} -** helps arginase, the final enzyme in urea cycle
- **Zn^{2+} –** assists alcohol dehyrogenase together with NAD+ to convert alcohol to aldehyde and carboxylic acid

Important coenzymes

- **FADH (Vit B_2)** flavin adenine dinucleotide – reducing agent in cellular respiration and donates electrons to the electron transport chain.
- **NADH (Vit B_3)** nicotinamide adenine dinucleotide – capable of carrying and transferring electrons and functions as oxidizing agent in redox reactions.
- **Quinone** – fully conjugated aromatic rings that acts as pigments in bacteria, fungi, and certain plants.
- **CoA (synthesized from Vit B_5)** – acyl group carriers

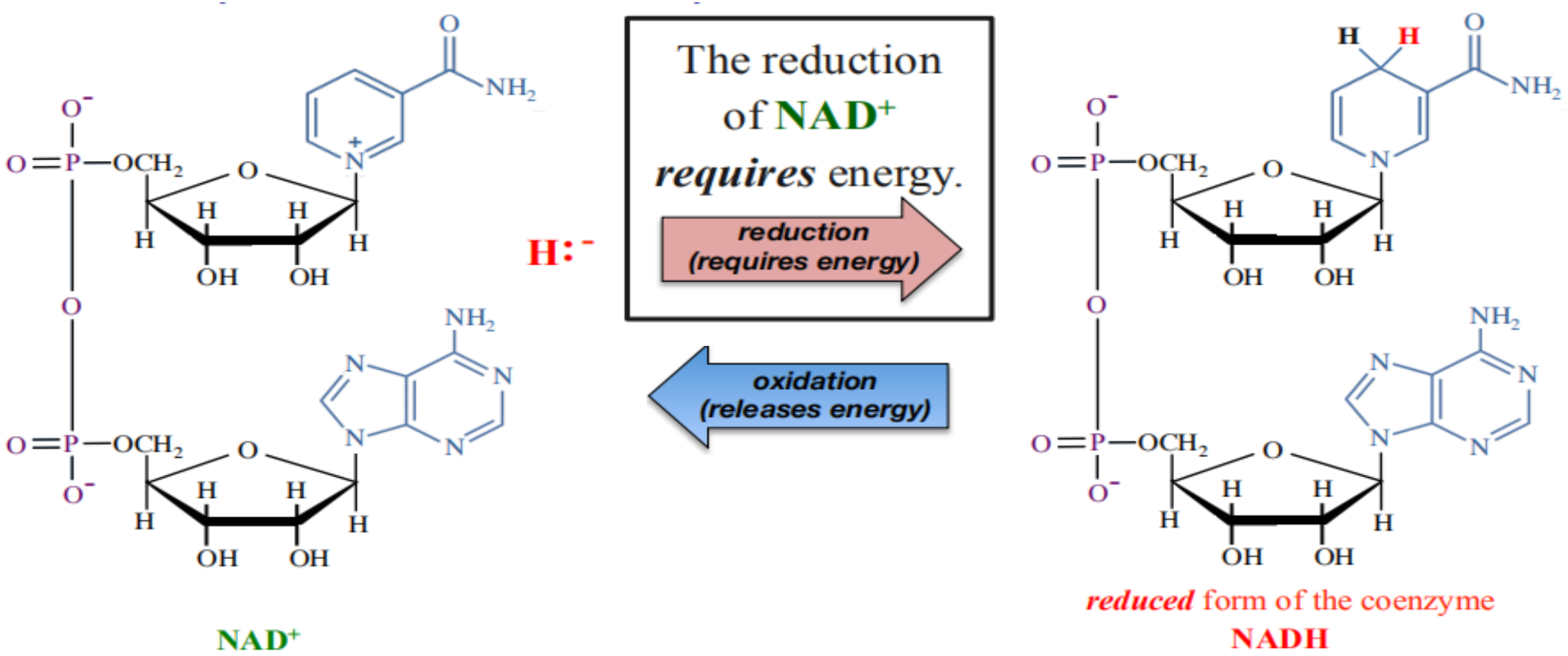
Other Coenzymes

- **Vitamin C** ascorbic acid – co-substrate with iron in formation of collagen in the body, helper of other vitamins (E and B_9) to remain active
- **TPP (Vit B_1)** thiamin pyrophosphate – needed in CAC – transfer of carboxyl group
- **Vitamin B_6 compounds (6) –** coenzymes for the transfer of amino groups
- **Biotin (Vit B_7 or Vit H)** – transfer of carboxyl group
- **THF (Vit B_9)** tetrahydrofolate – needed in methylation reactions
- **Cobalamin (Vit B_{12}) –** transfer of methyl group
- **Vitamins A, K**

Important coenzymes

NADH (Vit B) capable of carrying and transferring electrons and functions as **oxidizing agent** in redox reactions.

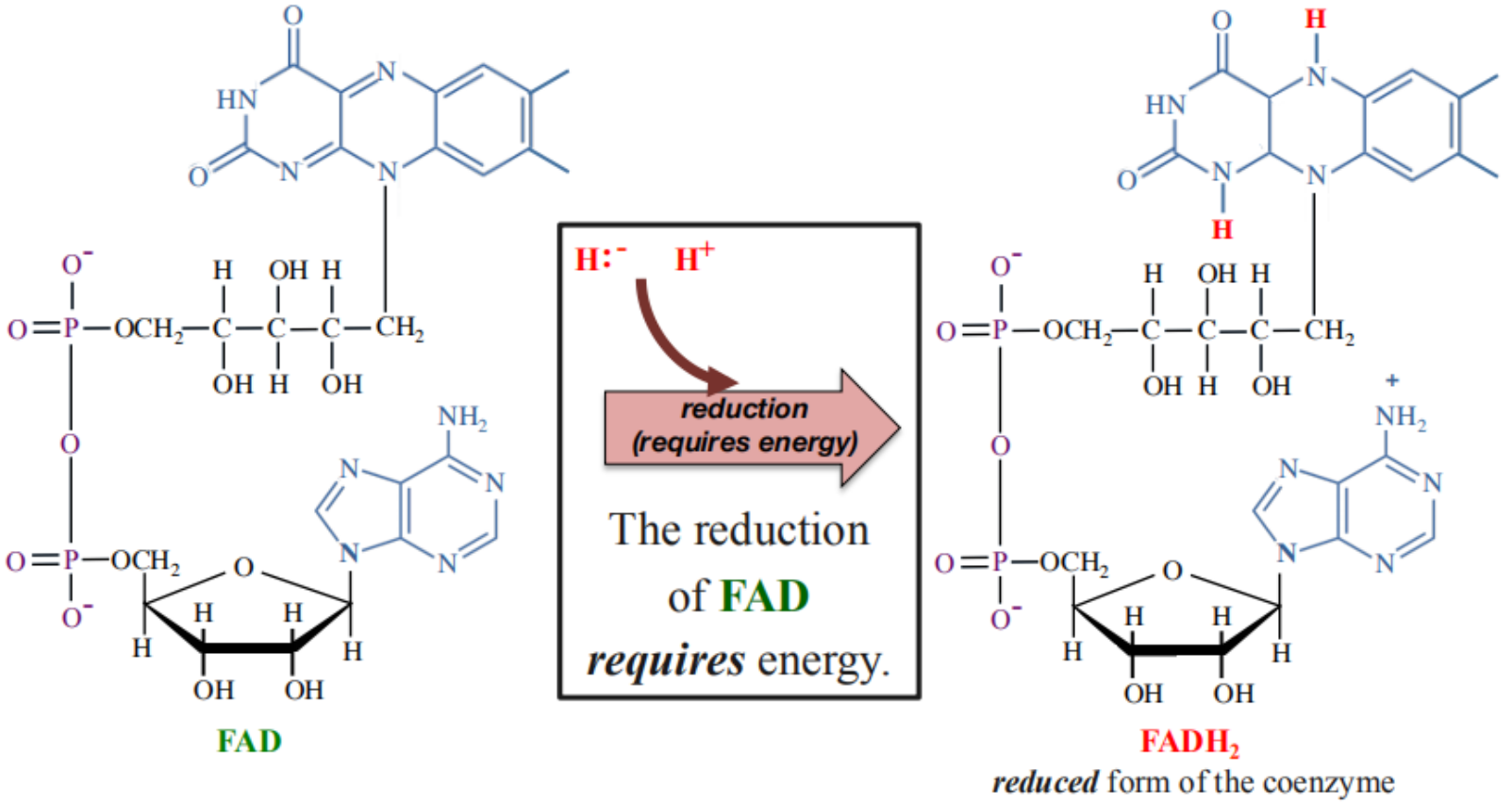
nicotinamide adenine dinucleotide –



Important coenzymes

FADH (Vit B_2) flavin adenine dinucleotide – **reducing agent** in cellular respiration and donates electrons to the electron transport chain.

When **FAD** *accepts* a hydride ion from another species (and an H^+ from solution), it is ***reduced*** to **$FADH_2$**



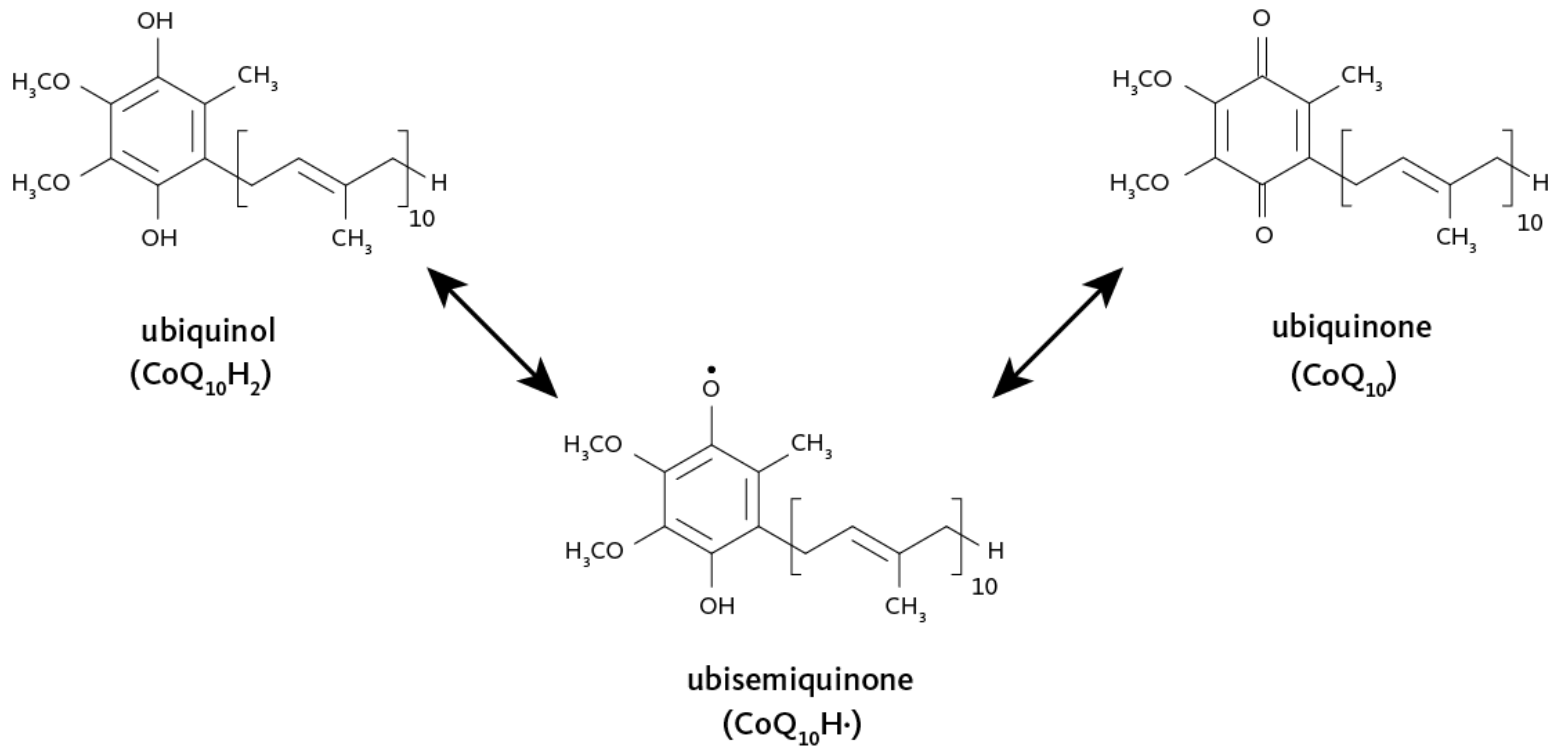
Important coenzymes

Coenzyme Q_{10} (ubiquinone)

– oil-soluble, vitamin-like substance

- Carries electrons from complexes in aerobic cellular respiration

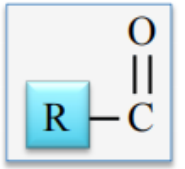
Figure 1. The Different Redox Forms of Coenzyme Q_{10}



Coenzyme Q_{10} exists in three oxidation states: the fully reduced ubiquinol form ($CoQ_{10}H_2$), the radical semiquinone intermediate ($CoQ_{10}H\cdot$), and the fully oxidized ubiquinone form (CoQ_{10}).

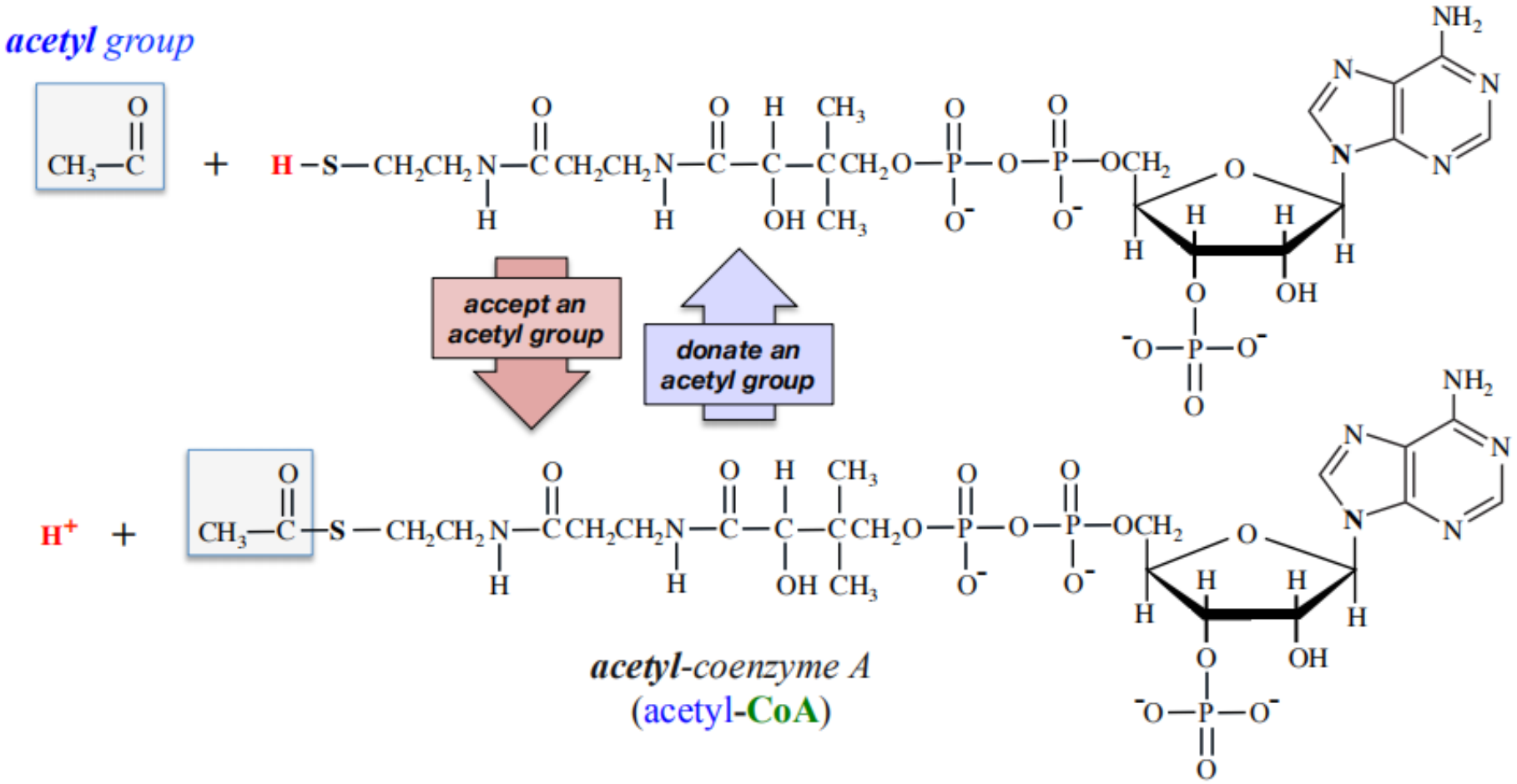
Important coenzymes

CoA (synthesized from Vit B_5) – acyl group carriers



an acyl group

Acetyl groups are donated and accepted by *coenzyme A*, as shown below:



Factors Affecting Enzyme Activity

Inhibitor

- Substances that reduce the rate of enzymatic reactions.
- The are usually specific and they work at low concentrations.
- They block the enzyme but they do not usually destroy it.
- Many drugs and poisons are inhibitors of enzymes in the nervous system.
- It can be reversible or irreversible

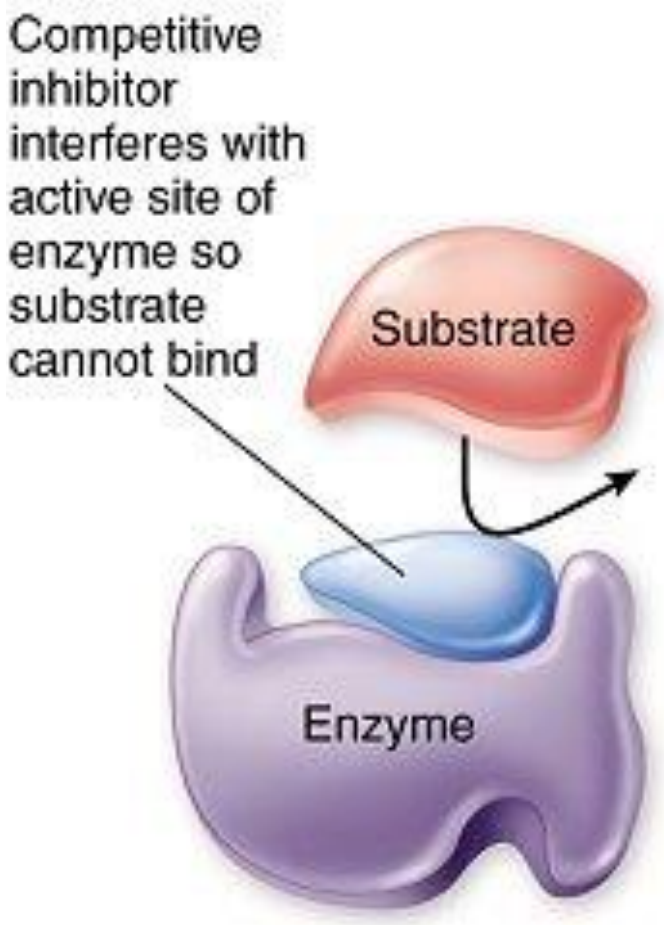
MODE OF ACTION

- Reversible Competitive inhibition
- Reversible Non-competitive inhibition
- Irreversible enzyme inhibition

Factors Affecting Enzyme Activity

Competitive Inhibition

- These compete with the substrate molecules for the active site.
- The inhibitor's action is proportional to its concentration.
- Resembles the substrate's structure closely.
- Effect can be reversed by increasing the concentration of substrate



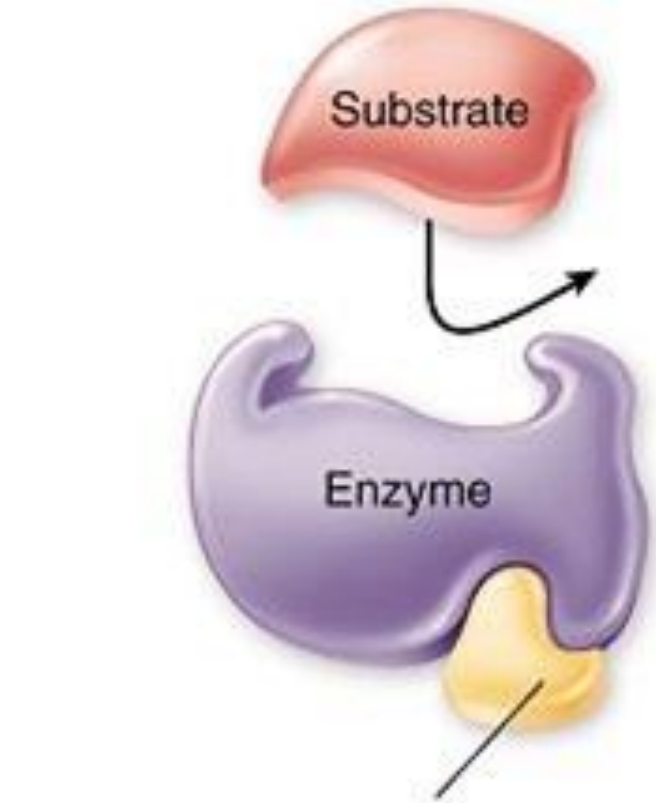
(a) Competitive inhibition

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Factors Affecting Enzyme Activity

Noncompetitive Inhibition

- Has a structure different from the substrate
- Distorts the shape of the enzyme when it is bound to it, thus enzyme can't bind the substrate.
- These are not influenced by the concentration of the substrate.
- It inhibits by binding irreversibly or reversibly to the enzyme but not at the active site.



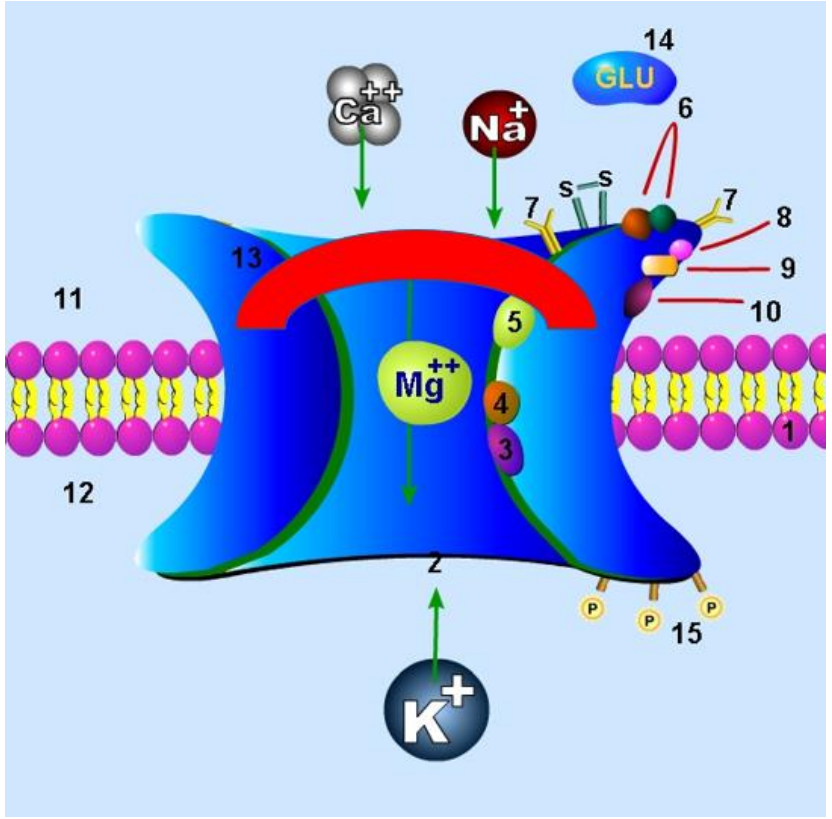
(b) Noncompetitive inhibition

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Factors Affecting Enzyme Activity

Uncompetitive Inhibition

- Has a structure different from the substrate
- Inhibition happens only when enzyme is already bound to the substrate.
- The concentration of substrate is reduced at the same time the rate of enzymatic action (substrate binds longer in the enzyme)
- It inhibits by binding irreversibly to the enzyme but not at the active site.



By Sazfar21 - Derivative of File:NMDA receptor.jpg by Blanca Piedrafita, CC BY-SA 2.5, https://commons.wikimedia.org/w/index.php?curid=77307446

Application of Inhibitors

Poisons: snake bite, plant alkaloids and nerve gases.

Medicine: antibiotics, sulphonamides, sedatives and stimulants

Examples:

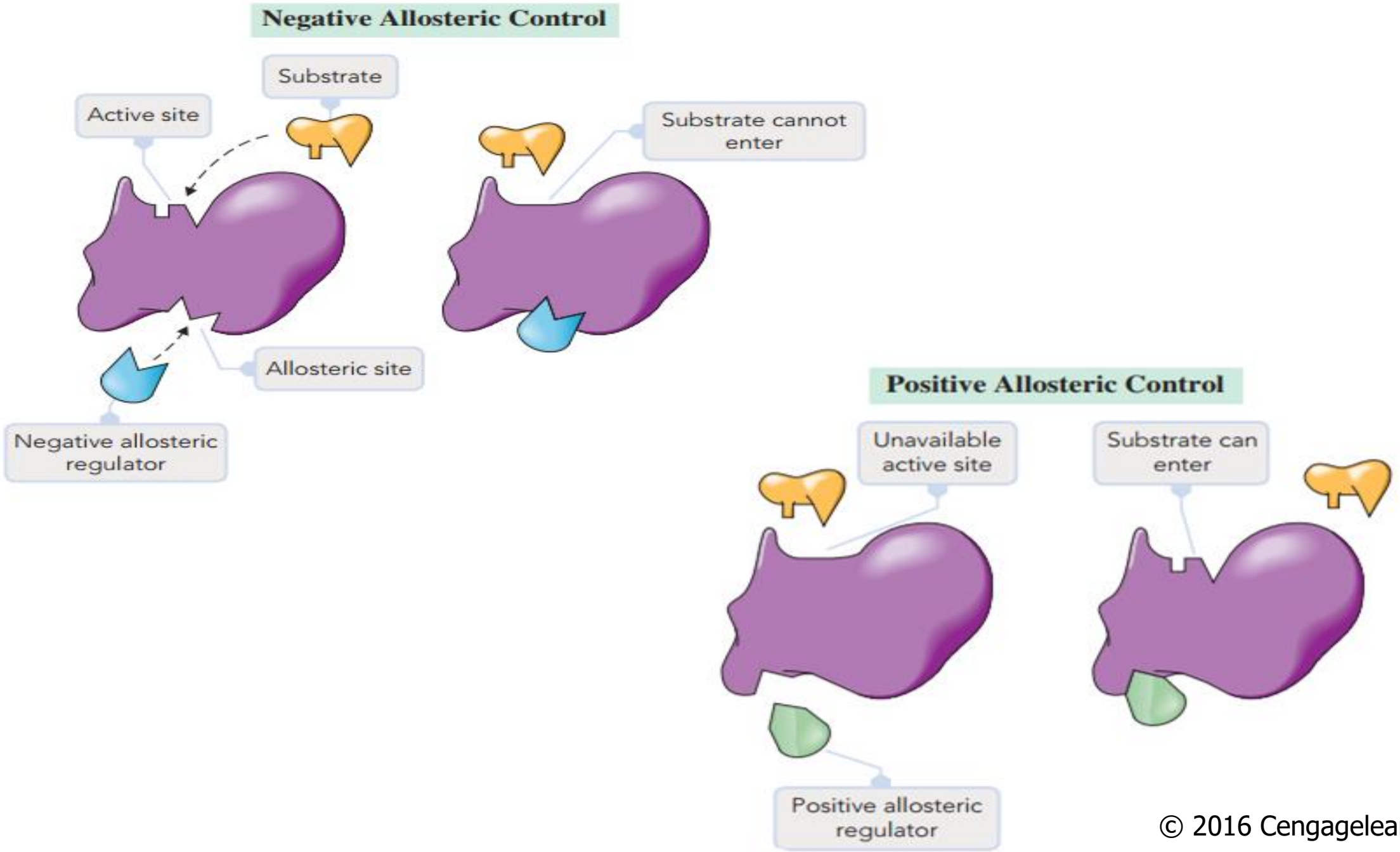
- **Aspirin** to inhibit the synthesis of prostaglandins which are at least partly responsible for the aches and pains of arthritis
- **Methotrexate** in cancer chemotherapy to semi-selectively inhibit DNA synthesis of malignant cells
- **Sulfa drugs** to inhibit the folic acid synthesis that is essential for the metabolism and growth of disease-causing bacteria

Application of Inhibitors

Allosteric Regulators

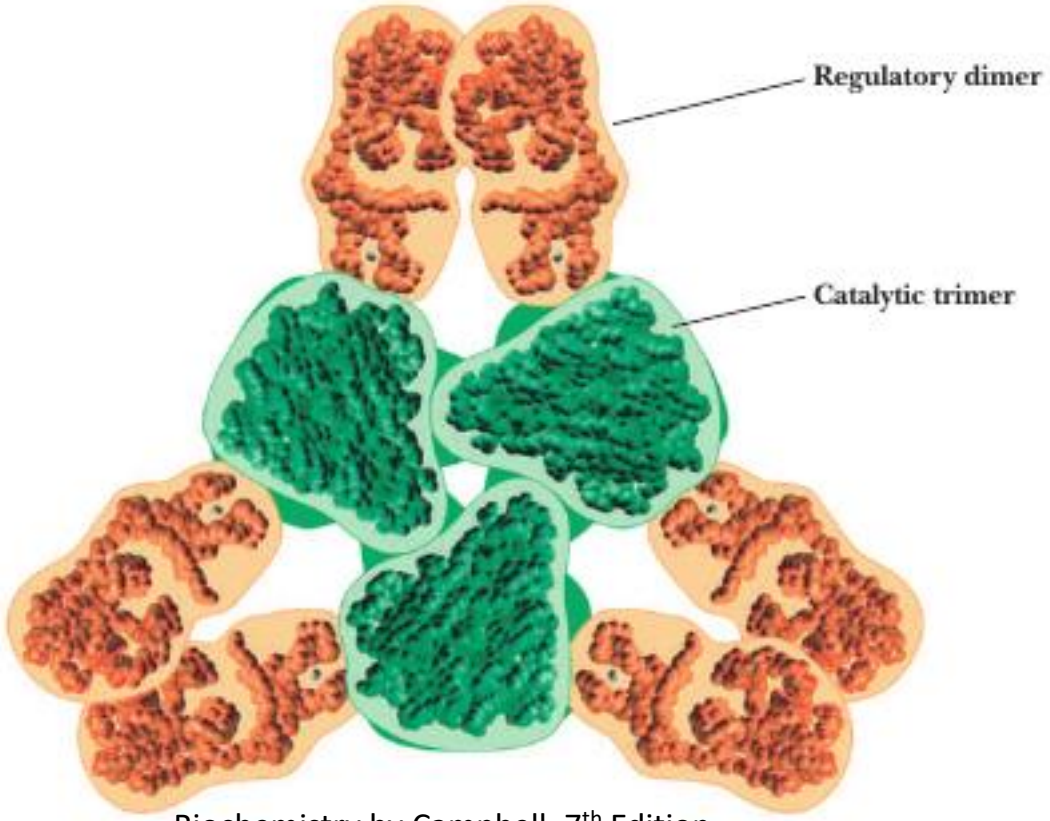
- **Negative regulator – inhibitor**
  - Substances that decrease the rate of enzymatic reaction
  - Changes the active site such that substrate is less readily accepted.

- **Positive regulator – activator**
  - Substances that increase the rate of enzymatic reaction
  - The shape of the active site changed such that it can more readily accept substrate
    - Examples: fructose-2,6-biphosphate which activates phosphofructokinase 1 and increases the rate of glycolysis in response to the hormone insulin.
    - Hexokinase-1
    - Glucokinase



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Example of Allosteric Enzyme



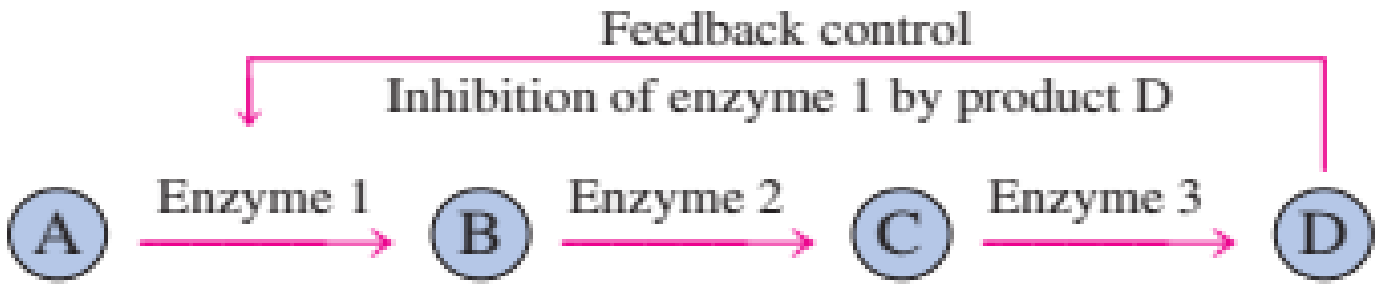
■ **FIGURE 7.3** Organization of aspartate transcarbamoylase, showing the two catalytic trimers and the three regulatory dimers.

Biochemistry by Campbell, 7th Edition

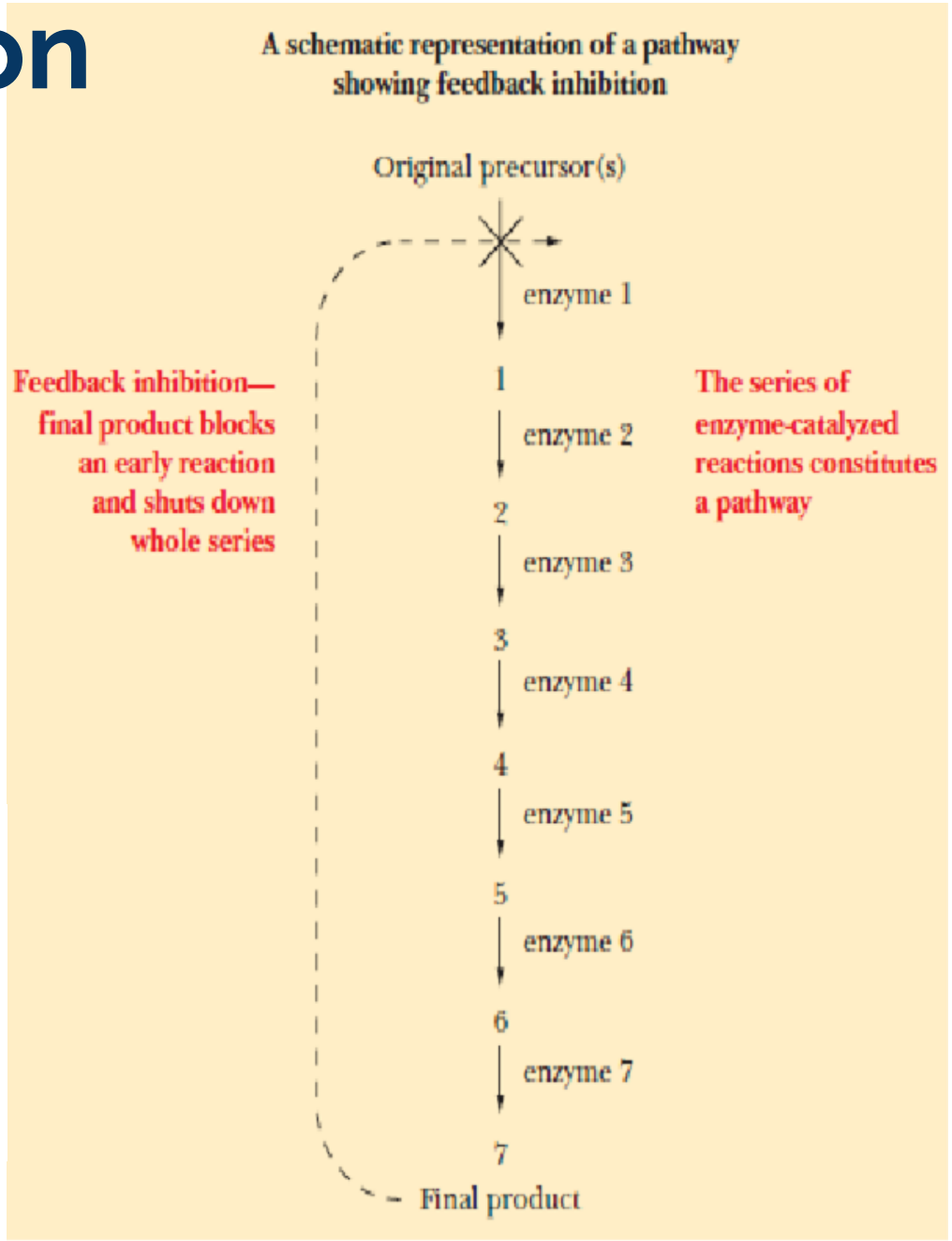
Inhibitor – cytidine triphosphate
Activator – adenosine triphosphate

Another Type of Regulation

- **Feedback Control**
  - Is a process in which activation or inhibition of the first reaction in a reaction sequence is controlled by a product of the reaction sequence



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REFERENCES

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- Brown, T. L., LeMay, H. E., Bursten, B. E., Murphy, C. J., & Woodward, P. M. (2012). Chemistry: The Central Science (12th ed.). Pearson Prentice Hall.
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